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Alzheimer's disease diagnosis using deep learning techniques: datasets, challenges, research gaps and future directions

Asifa Nazir¹ · Assif Assad¹ · Ahsan Hussain¹ · Mandeep Singh²

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Abstract

Alzheimer's disease (AD) is a condition characterized by the degeneration of brain cells, leading to the development of dementia. Symptoms of dementia include memory loss, communication difficulties, impaired reasoning, and personality changes, often deteriorating as the disease advances. As per the statistics, around 6.9 million individuals in the United States are diagnosed with AD. Approximately two-thirds of Americans with Alzheimer's are female. Of the total population affected, 4.2 million are women, while 2.7 million are men aged 65 and older in the U.S., constituting 11% of women and 9% of men within this age group. While treatment options for AD are available, they primarily aim to address symptoms rather than providing a cure or slowing down the progression of the disease. Several neural network scans play crucial roles in medical diagnostics, including "Magnetic Resonance Imaging (MRI)" and "Positron Emission Tomography (PET)". However, these techniques often involve manual examination, resulting in drawbacks such as slow processing and the risk of human error. This study aims to demonstrate how Artificial Intelligence (AI) techniques, including computer vision, Machine Learning (ML), and Deep Learning

(DL), can precisely diagnose the early stages of AD, potentially delaying or preventing disease progression. DL algorithms, known for their ability to handle vast amounts of data and extract relevant features, allow the detection of treatable symptoms of the disease before it reaches irreversible stages. The study begins with an overview of AD and the prevailing methodologies utilized for its early detection. It delves into examining diverse DL techniques in scrutinizing clinical data to identify the disease in its early stages. Further, the study explores various publicly accessible datasets, addressing associated challenges and proposing potential future research directions. A significant contribution of this research lies in introducing holography microscopic medical imaging as a novel approach to AD diagnosis, an area previously unexplored by researchers. The discussion section thoroughly explores different interpretations and implications arising from the conducted study. The second last section addresses ongoing research obstacles and looks at potential avenues for future studies. Ultimately, the study concludes by presenting its findings and considering their implications.

Keywords Alzheimer's disease (AD) · Deep learning (DL) · Early diagnosis · Convolutional neural networks (CNNs) · Recurrent neural networks (RNNs)

✉ Asifa Nazir
asifa.nazir@iust.ac.in

Assif Assad
assif.assad@islamicuniversity.edu.in

Ahsan Hussain
ahsan@iust.ac.in

Mandeep Singh
mandeep@iust.ac.in

¹ Department of Computer Science & Engineering, Islamic University of Science & Technology, Awantipora, Jammu & Kashmir, India

² Department of Physics, Islamic University of Science & Technology, Awantipora, Jammu & Kashmir, India

1 Introduction

Alzheimer's disease (AD) is a progressive neurological disease characterized by the gradual degeneration of nerve cells. Its hallmark features include the presence of neuritic plaques [1] and neurofibrillary tangles [2], which result from the accumulation of Amyloid-Beta peptides ($A\beta$), primarily in the medial temporal lobe and neocortical structures, ultimately leading to cell death [3]. The term "AD" was coined by Emil Kraepelin in the eighth edition of his psychiatry handbook, marking its official recognition as a

medical condition. During his examination of a patient's brain who had exhibited symptoms of memory loss and personality shifts before death, Alois Alzheimer observed the presence of amyloid plaques and a substantial decline in neuronal count. Various factors contribute to the progressive deterioration of cognitive functions in the brain, including intoxications, infections, abnormalities in the pulmonary and circulatory systems, reduced oxygen supply to the brain, inadequate nutrient intake, vitamin B12 deficiency, and related factors [4]. In the United States, an estimated 6.9 million people are living with Alzheimer's dementia. Among the total affected population, 4.2 million are women, while 2.7 million are men aged 65 and older in the U.S., representing 11% of women and 9% of men in this age group [5]. Figure 1(a) presents statistical details regarding AD prevalence in the United States followed by Figure 1(b) providing statistical data related to deaths due to AD.

Advancements in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL) and computer vision techniques have significantly improved early disease diagnosis. These technologies provide novel research opportunities for accurate and precise detection, surpassing the diagnostic capabilities of traditional human methods. Utilizing AI and ML, medical practitioners can now detect diseases at their early stages with improved precision [6]. Many studies suggest the use of DL algorithms like Convolutional Neural Networks (CNNs) [7] for AD diagnosis [8–28]. However, comparing classification results across these studies is difficult due to variations in factors such as participant numbers, image processing methods, cross-validation procedures, and reported evaluation metrics, making explicit comparisons challenging. Therefore, determining the optimal approach is complex, and the extent of its generalization ability remains to be undetermined. Traditional diagnostic techniques like manual examination of medical images or evaluation of observable symptoms in healthcare frequently encounter limitations because they depend on subjective interpretation and are prone to human error. These strategies can be time-consuming and may not accurately detect minute or early-stage diseases. DL algorithms are proficient in efficiently processing vast amounts of data and identifying subtle patterns that may signify the presence of disease. This ability increases diagnostic accuracy by facilitating earlier detection and intervention, significantly improving patient results. By utilizing diverse datasets, DL algorithms improve their generalization across various subjects and conditions, allowing the trained detection of subtle patterns for earlier and more precise disease diagnosis. DL and computer vision techniques such as CNNs and advanced image recognition algorithms like Recurrent Neural Networks (RNNs) [29], Long Short-Term Memory (LSTM) [30], Gated Recurrent Unit (GRU) [31], Generative

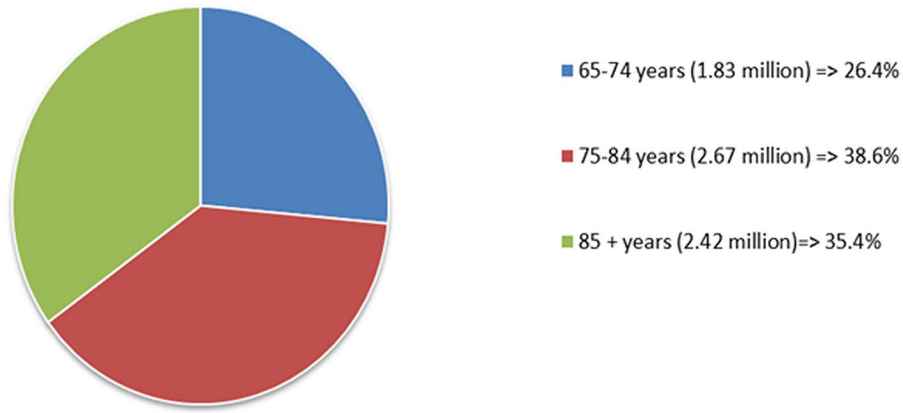
Adversarial Networks (GANs) [32], transformer-based architectures [33], AutoEncoder (AE) [34] etc., present a promising alternative by automating medical data analysis. Figure 2 depicts the distribution of chronic disease conditions, showcasing their respective percentages.

Various cutting-edge imaging techniques are employed to detect AD, driven by implementing DL methodologies. These imaging modalities include “Magnetic Resonance Imaging (MRI),” “Positron Emission Tomography (PET),” “Single-Photon Emission Computed Tomography (SPECT),” “Computed Tomography (CT),” “Functional MRI (fMRI)” and “Diffusion Tensor Imaging (DTI)”. Each technique provides a distinct perspective on the intricate mechanisms underlying AD pathology. MRI [35], in particular, stands out for its ability to facilitate detailed exploration of the pathological changes associated with AD. The integration of neuroimaging data with AI techniques is transforming AD detection, offering opportunities for personalized diagnosis and tracking disease progression. Incorporating predetermined features, such as metrics derived from image processing pipelines, including regional and voxel-based data, and combining them with various classifiers like Support Vector Machine (SVM) [36] or random forest can significantly enrich detection accuracy. These cutting-edge methodologies, referred to as traditional ML approaches, lead the way in advancing AD diagnosis. DL has become a rapidly evolving technology, achieving significant advancement in medical imaging [37], [38], [39]. Its growing popularity has attracted much attention, as it has successfully classified images with and without disease symptoms [40]. DL identifies low-level features such as lines, edges, and dots and high-level features encompassing tumor shapes, sizes, and textures in the case of malignancy diagnosis [41]. When combined with various imaging modalities, DL algorithms show significant potential for early AD detection by integrating clinical data with advanced computational models.

This fusion of DL and clinical data allows the development of precise and efficient diagnostic tools, empowering healthcare professionals to effectively identify the early stages of AD. DL holds significant promise for improving diagnosing AD's accuracy, speed, and cost-effectiveness by extracting valuable patterns and features from intricate datasets. This progress can lead to better patient results and more efficient resource allocation in healthcare settings. In DL, using entirely independent test sets is particularly vital compared to conventional ML, given the improved flexibility offered by various model architectures and multiple hyper-parameter choices (such as learning rate, number of epochs, etc.) [42]. DL models excel in accurately identifying diseases through various imaging modalities, but they often need help with issues like overfitting and underfitting.

(a)

Number and Ages of People 65 or Older with Alzheimer's Dementia, 2024 (Total = 6.9 million)



(b)

STATISTICAL DATA RELATED TO DEATHS DUE TO AD

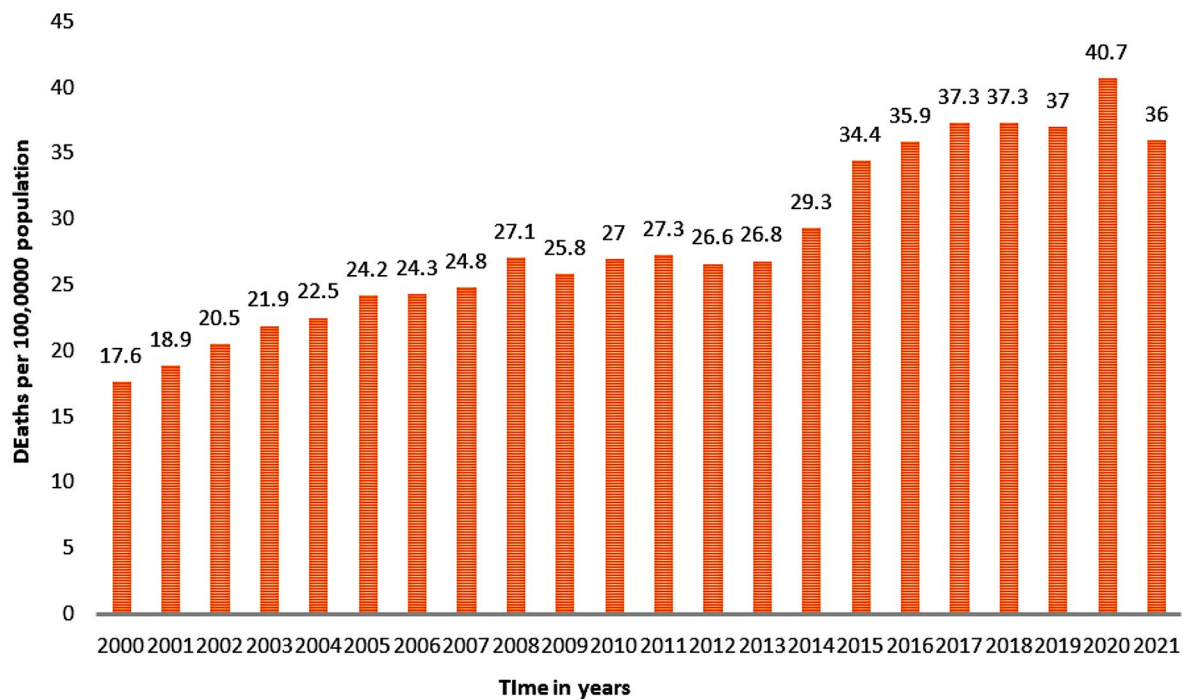


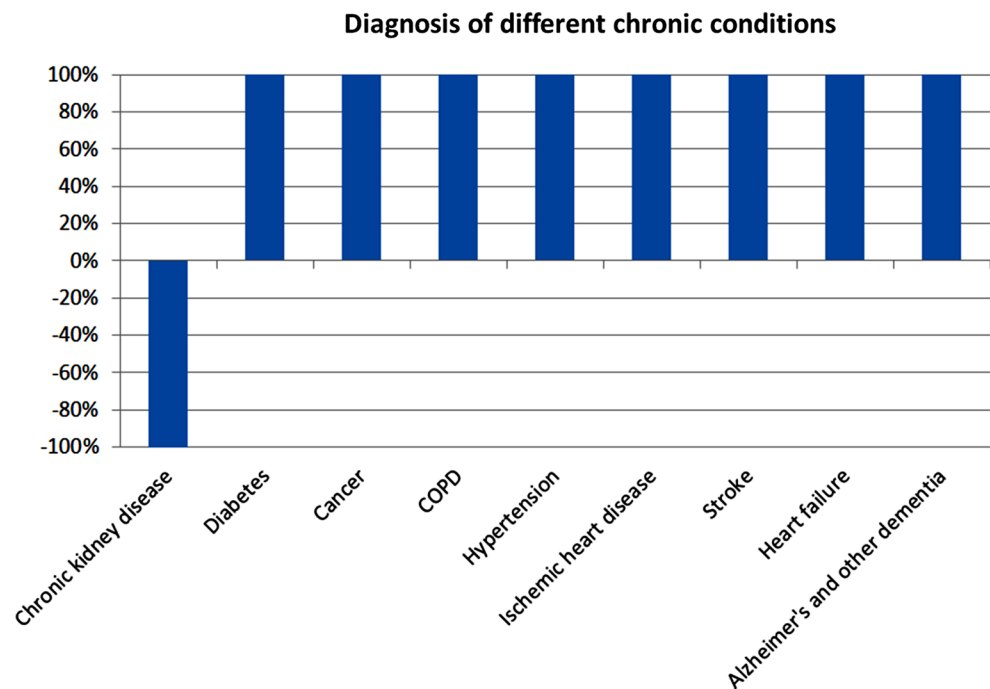
Fig. 1 Statistical data corresponding to AD. (a) Statistical details regarding AD prevalence in the United States. (b) AD mortality rate in the United States from 2000 to 2019 (per 100,000 population) [<https://www.statista.com/statistics/452945/mortality-rate-of-alzheimers-patients-in-the-us/>]

Overfitting arises when a model learns the training data closely, capturing noise specific to the training set, thereby failing to generalize to new data. This occurs when the model is more complex than the available training data. In contrast, underfitting occurs when a model is overly simplistic, leading to poor performance on training and unseen

data due to a lack of complexity or training data. Methods like regularization, dropout, and early stopping, merged with increased training data diversity or volume, mitigate these concerns [43].

The proposed research study contributes by offering an extensive exploration of AD and investigates cutting-edge

Fig. 2 Percentage of different chronic disease conditions [5]



techniques in computer vision for its early diagnosis, aimed at halting its progression to irreversible stages. It systematically explores various DL architectures with significant potential for early disease detection. Essential classification methods in AD diagnosis, including end-to-end CNNs, 2D slice-level CNNs, 3D patch-level CNNs, ROI-based CNNs, and 3D subject-level CNNs, are analyzed along with their respective benefits and limitations. Future research highlights the novelty of the study by identifying the significance of holography microscopic medical imaging in early disease diagnosis, an aspect unexplored in any AD study. Moreover, the study addresses challenges and data leakage problems (like transfer learning bias) and presents various AD-related datasets for research in this area. It outlines the challenges and future research directions, identifying the novelty of this study. *The paper is structured as follows: Sect. 1 introduces AD diagnosis using DL techniques, including the paper searching strategy and motivation behind the study. Then, Section 2 provides an overview of the current state-of-the-art, followed by Sect. 3, discussing various DL architectures. Section 4 examines AD classification areas and data leakage issues in clinical algorithm evaluation. Section 5, introduces the concept of transfer learning as a revolutionary approach to AD diagnosis. Section 6 explores diverse imaging modalities and neural network scans for AD detection. Section 7 covers datasets and databases related to AD. Section 8 presents a novel approach to integrating holography microscopic medical imaging with DL for disease diagnosis. Section 9 highlights challenges, research gaps, and future directions. Sections 10 and 11 cover the discussion and conclusion, respectively.*

1.1 Paper searching strategy

In this research, the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [44] methodology is the primary approach for conducting systematic reviews and meta-analyses of research studies. Peer-reviewed articles, conference papers, and reviews were accessed using academic databases such as PubMed, IEEE Xplore, and Google Scholar. The search results are refined based on publication date, document type, and relevance criteria. Following the PRISMA strategy, 300 papers were identified as relevant to this study. After careful screening to remove duplicates and irrelevant articles, only 200 are found relevant. Ultimately, for this review, 177 research publications were included. Figure 3 shows the use of the PRISMA strategy in collecting research publications for this study. The paper search strategy for this research entails identifying pertinent keywords related to AD, DL, early detection, CNN, RNN, vision transformers [45], explainable AI [46], and critical review. Figure 4 displays the search string utilized to identify relevant papers.

1.2 Motivation of the study

The motivation behind this study stems from the pressing need to improve early detection methods for AD. Early diagnosis is critical in AD management as it allows for timely intervention, which can significantly impact patient results. By recognizing AD during its initial phases, individuals can access treatments and interventions early, potentially preventing progression to irreversible stages. Further, early

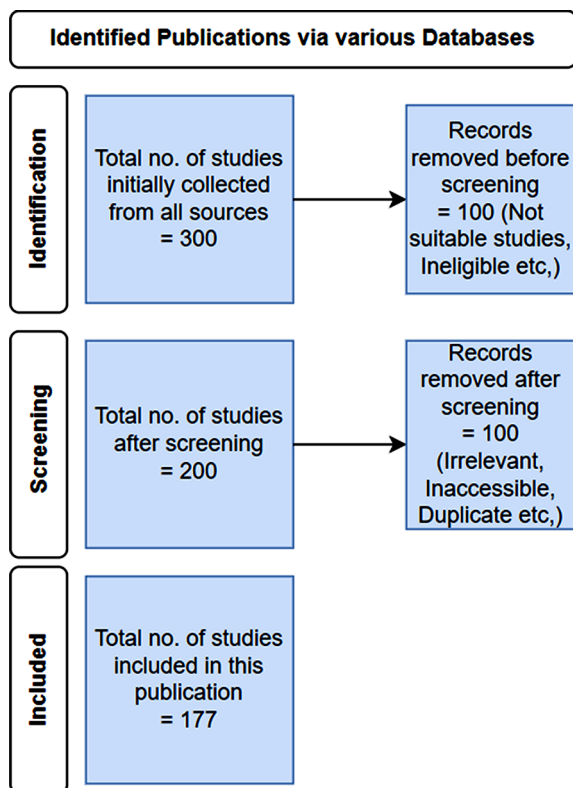


Fig. 3 PRISMA strategy followed in this study

detection enables individuals to participate in clinical trials for potential therapies, fostering the development of novel treatment approaches. Thus, this study's primary motivation is to improve early AD diagnosis techniques, ultimately improving patient care and outcomes. Figure 5 provides an abstract view of this study, encapsulating its aim and scope. The following section introduces state-of-the-art techniques for diagnosing AD via DL methods.

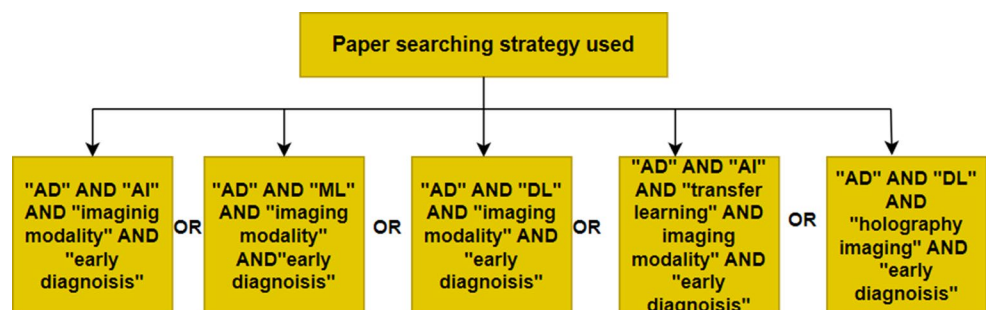
2 State-of-the-art

State-of-the-art research on the early diagnosis of AD with the integration of DL strategies with clinical data has emerged as a critical area of this study. This review analyzes the current advancements, methodologies, and challenges

associated with leveraging DL techniques for early diagnosis. Through integrating clinical data and advanced computational models, DL has demonstrated encouraging results in improving the precision and effectiveness of AD detection. The comprehensive review evaluates current methodologies, highlighting both their benefits and limitations. Many researchers conducted classification experiments across different stages of the disease to improve diagnosis. Utilizing ML and DL techniques, they analyze medical data to identify patterns indicative of AD, advancing understanding and diagnostic accuracy. The objective is to improve the development of treatment strategies with increased effectiveness. Notably, many studies encounter biased evaluation results due to data leakage issues [47]. A comprehensive examination of image data from the selected studies reveals various associated challenges. Proficiency in DL and **medical image processing methods like MRI and CT scans** are vital for crafting robust DL methodologies for AD classification. New entrants in this area may find it challenging to acquire these skills; researchers can refer to [48] for DL insights and [49] for MRI processing expertise.

In recent years, DL has seen considerable adoption in medical image analysis, showcasing notable accomplishments and attracting researchers needing to analyze various medical images. DL models offer superior accuracy and efficiency in detecting diseases like AD compared to traditional ML methods. These benefits motivate researchers to utilize DL for early disease detection and ongoing monitoring across various medical imaging modalities. This study extensively discusses various DL architectures outlined in the existing literature and thoroughly investigates potential neural scans for detecting AD. Utilizing **CNNs**, the [50] focuses on early disease diagnosis using MRI image datasets. The study has used **MobileNet architecture**, renowned for its lightweight design and depthwise separable convolutions, resulting in an impressive **accuracy of 96.22%** in multi-classification tasks. [51] employs an ensemble technique to detect AD, **integrating classifiers like Bi-GRU, Multi-Layer Perceptron (MLP) [52], and Quantum Neural Network (QDNN) [53]**. These classifiers are amalgamated to form the ensemble classifier, with the Bi-GRU model trained using outputs from MLP and QDNN. **Conditional**

Fig. 4 Paper searching strategy



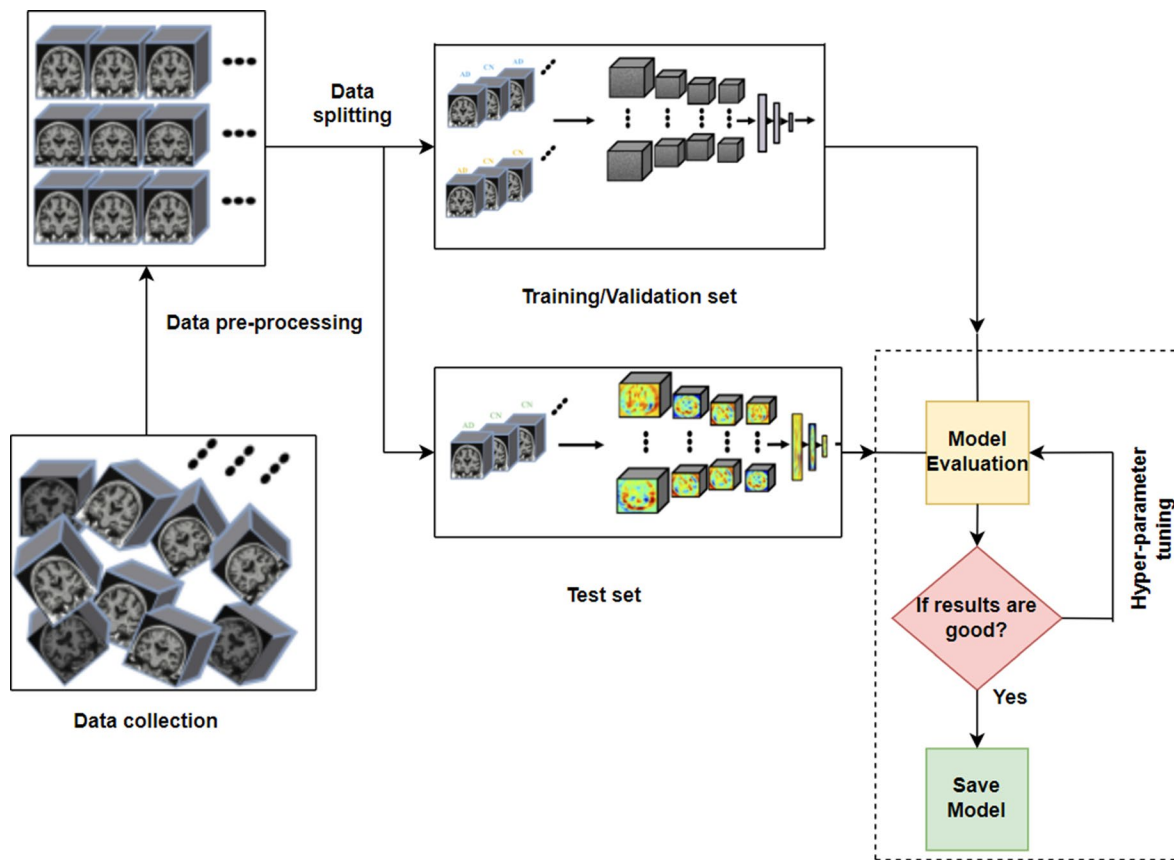


Fig. 5 Abstract view of the study

progressive GAN (cProGAN) is used by [54] to enhance AD classification using MRI data. By adjusting to feedback from a discriminator trained on labeled data, cProGAN generates synthetic data and selects samples based on the distributions of original data mapped into the Linear Discriminator Analysis (LDA) space. These generated samples aid in training CNNs, consequently improving classification accuracy. The GAN model utilized in their study is a modified version of conditional GANs, incorporating noise vectors and label vectors to create synthetic MRI samples.

One of the studies by [55] utilized a Vision Transformer (ViT) to classify AD stages using MRI and PET data. These modalities are fused through preprocessing and image registration before training the ViT model, resulting in high accuracy rates for AD identification. ViTs, with their robustness, efficiency, generalization, and transfer learning advantages over CNNs, emerge as suitable for processing multimodal neuroimaging data in AD diagnosis. ViTs improve disease determination accuracy by effectively processing multimodal medical imaging data, such as MRI and PET scans, leveraging their robustness and generalization capabilities. Their efficient representation of learning mechanisms and ability to capture complex patterns contribute to improved disease classification performance. A combinational study

of CNN and RNN is employed to analyze structural MRI images to diagnose AD in [56]. It addresses issues in previous approaches by learning spatial and longitudinal features and disease classification. Analyzed with longitudinal T1-weighted MRI data from the ADNI database, the proposed method achieves promising classification accuracies of 91.33% for AD vs. NC and 71.71% for pMCI vs. sMCI, indicating its effectiveness in longitudinal MR image analysis for AD diagnosis. The following section explores different DL architectures, including supervised, unsupervised, and reinforcement learning algorithms.

3 Deep learning architectures

AI, ML, and DL have gained widespread attention across healthcare, agriculture, business, education, banking, and IT sectors. These terms are ubiquitous, reminiscing their extensive use and impact across various fields. AI, ML, and DL are frequently employed and sometimes used interchangeably, but it's vital to comprehend their nuanced disparities. AI enables machines, including robots, to mimic human-like behaviors and intelligence. ML, a subset of AI, allows machines to learn from historical data using feature

extraction algorithms. In contrast, DL, another subset of ML, employs deep ANN with multiple layers to learn from vast datasets, drawing inspiration from the intricate workings of the human brain. These Deep Neural Networks (DNNs) [57] comprise initial, intermediate (hidden), and output layers, progressively representing information sophisticatedly. The term “deep” signifies networks containing more than two fully connected layers. In classification tasks, one can utilize either a shallow neural network with a single hidden layer or a DNN with multiple hidden layers, offering flexibility in model architecture. Computer vision, a subset of AI encompassing ML and DL, has revolutionized how machines perceive and interpret visual data, allowing tasks like facial recognition [58], object detection [59], and autonomous navigation [60]. In scientific computing, DL algorithms such as CNNs, LSTMs, and GANs play pivotal roles across domains like computer vision, Natural Language Processing (NLP) [61], and generative modeling, contributing to substantial advancements in these fields. Figure 6 illustrates the classification of DL architectures, categorizing them into supervised, unsupervised, and reinforcement learning paradigms.

3.1 Supervised learning algorithms

Supervised learning algorithms involve training a model with labeled data to establish relationships between input features and output labels. The model adjusts its parameters to minimize the disparity between its predictions and the actual labels using a predefined loss function. These models

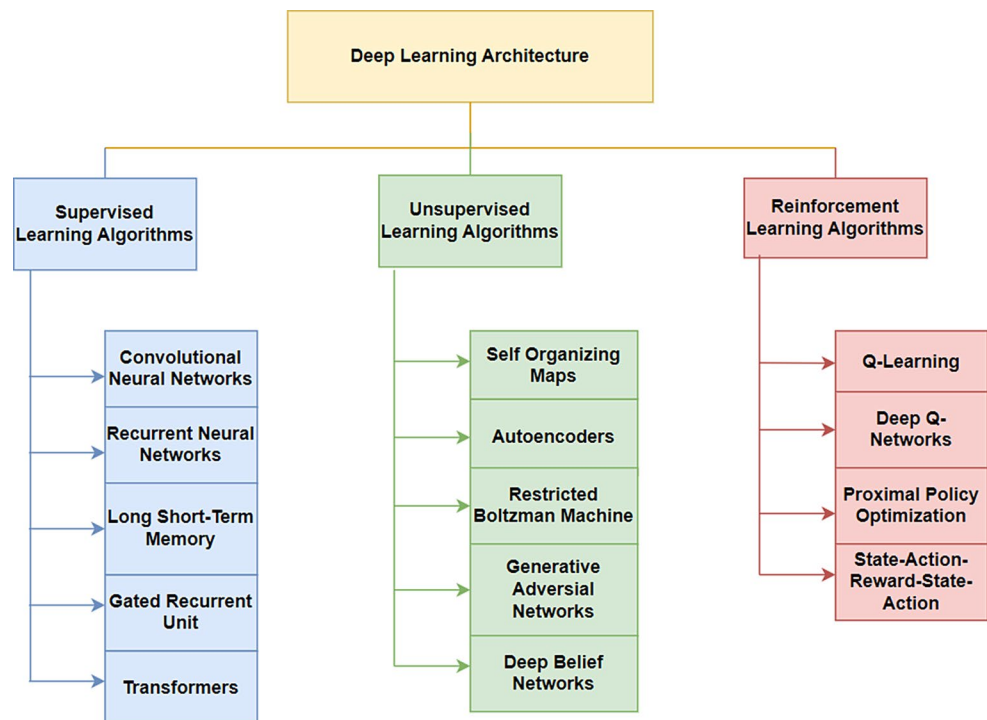
aim to generalize its learned patterns and accurately predict labels for new data instances [62].

3.1.1 Convolutional neural networks

CNNs, commonly known as ConvNets, are sophisticated architectures comprising multiple layers explicitly designed for efficient image processing and accurate object detection tasks. In 1988, Yann LeCun achieved a groundbreaking advancement with the development of LeNet, marking the inception of the first-ever CNN. This achievement stands as a remarkable milestone in the realm of DL. Designed meticulously, LeNet showcased exceptional proficiency in recognizing and classifying characters, including ZIP codes and digits, signifying a noteworthy advancement in pattern recognition. These networks have gained immense popularity for their broad utilization in processing and analyzing medical images, catching anomalies in time series predictions, and various other applications. Their widespread usage across different fields highlights remarkable capabilities and effectiveness across multiple research domains [63].

CNNs consist of several layers: input, convolutional (operating activation functions like ReLU), pooling, fully connected, and output layer. Each layer performs distinct tasks such as feature extraction, improving non-linearity, down-sampling, and generating predictions. These networks excel in pattern recognition, image classification, and DL analysis. The input layer of a neural network represents grayscale images (black and white) or colored images through Red, Green, and Blue (RGB) channels, with pixel

Fig. 6 Classification of DL architectures



values usually ranging from 0 to 255. The convolution layer filters the input image, effectively extracting or detecting its distinctive features. This process generates a feature map crucial for classifying the input image. Strides in CNNs control how the filter moves across the input image during convolution, adjusting its shift between operations. This extracts features at various resolutions and downsamples the output, enhancing computational efficiency without compromising feature capture. After the convolution operation, the resulting feature map is smaller than the input image, reflecting spatial size reduction. Increasing the stride value decreases the feature map size, showing an inverse correlation. The Pooling layer function is to reduce feature map dimensions for faster computations. A flattening layer transforms multi-dimensional feature maps into a one-dimensional vector for classification. The fully connected layer connects the information extracted from preceding layers to the output layer, facilitating precise predictions [64]. Figure 7 illustrates various operations commonly used in CNNs, such as convolution, pooling, activation functions, etc.

3.1.2 Recurrent neural network

RNNs, a type of neural network, utilizes recurrent connections to allow bidirectional information flow across preceding and current layers, making them distinctive from other neural network architectures. This unique design empowers RNNs to process sequential data efficiently [65]. At the core of RNNs is the “Hidden layer,” which retains and stores information about sequences, maintaining crucial temporal dependencies and encoding meaningful patterns. RNNs retain past input knowledge to model real-time problems, capturing context and addressing dynamic sequences accurately. RNNs form the foundational architecture for various advanced DL models, enabling sophisticated sequential data modeling across diverse applications like image captioning, time series analysis, NLP, handwriting recognition, and machine translation. They integrate current and previous inputs to capture temporal dependencies and make

informed predictions, making them a type of Feedforward Neural Network (FFNN) with memory and feedback loops. RNNs operate consistently across input data points, effectively capturing dependencies and patterns in sequential data. LSTM and GRU are specialized versions of RNNs, each designed to tackle specific issues encountered during processing sequential data. These variants excel in capturing long-term dependencies and mitigating the vanishing gradient problem often faced in conventional RNN systems. The general structure of RNNs is illustrated in Figure 8, while LSTM and GRU represent specific variations of the RNN architecture.

3.1.2.1 Long short term memory LSTM, a specialized type of RNN, deals with issues like vanishing or exploding gradients while effectively capturing long-term patterns in sequential data. It introduces a memory unit called a cell to handle short-term memory limitations, updating itself to maintain and process long-term patterns. In LSTM, decision-making involves integrating current input, prior output, and memory cell state, enabling output generation and memory adjustment. The architecture includes three gates—the forget gate, input gate, and output gate—to control data flow, setting it apart from traditional RNNs [30]. Figure 9 illustrates how LSTM processes sequential data, followed by Figure 10 detailing the operations within LSTM.

3.1.2.2 Gate recurrent unit To address the constraints of conventional RNNs, the GRU network emerges as a resilient and effective solution, providing enhanced learning abilities and superior information retention. GRU networks are typically equipped with two gates, the reset gate and the update gate, which play a pivotal role in managing information flow. In contrast to LSTM, where information gets stored in an internal cell, GRU integrates information into the hidden state, facilitating its propagation to subsequent GRU units. This feature enables GRU to effectively preserve and transmit information across the network, contrast-

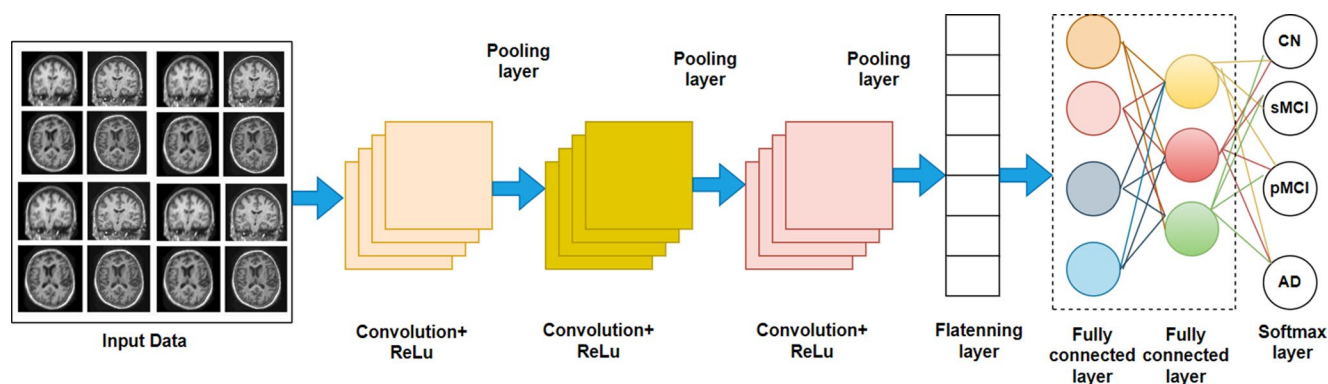
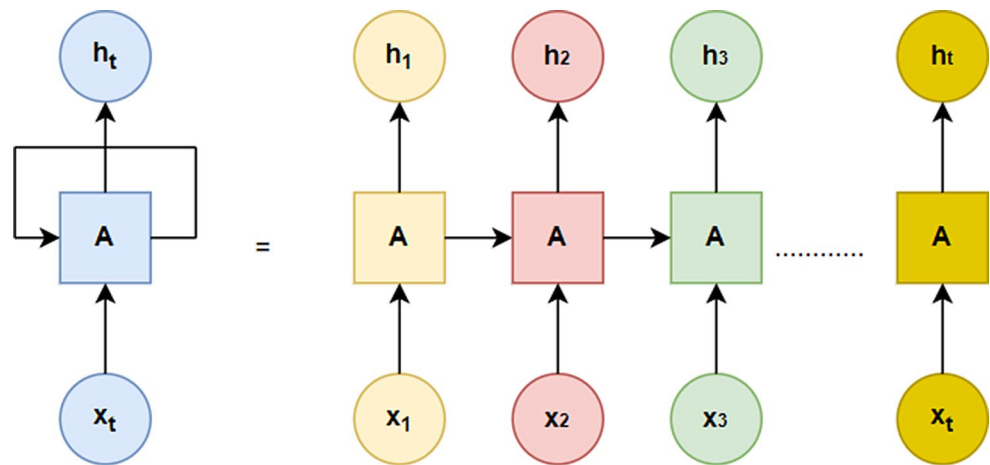
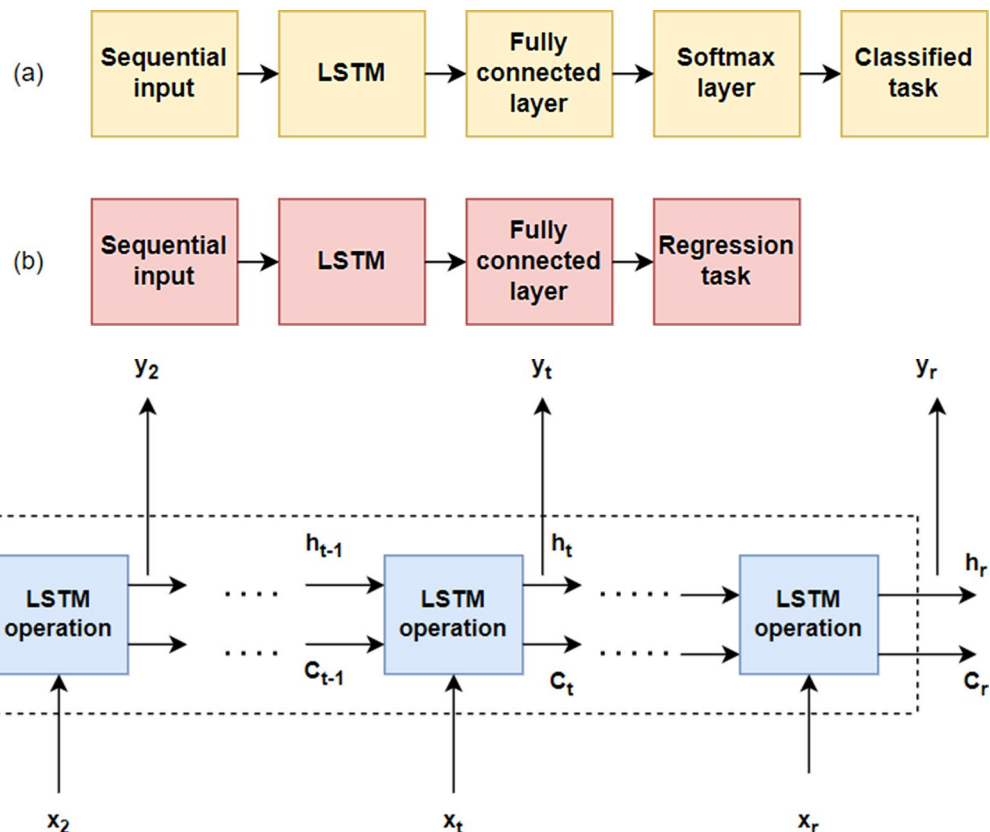


Fig. 7 Various operations commonly used in CNNs

Fig. 8 Structure of RNNs**Fig. 9** Sequential data processing using LSTM**Fig. 10** Operations inside LSTM network

ing it from other recurrent models. While sharing structural similarities with LSTM, GRU lacks a distinct cell state and relies on update and reset gates to regulate information flow [66].

3.1.3 Transformers

The transformer architecture proposed by [67] relies solely on attention mechanisms, eliminating recurrence

and convolutions. Experiments show superior translation quality, faster training, and improved parallelization. For instance, it achieves a Bilingual Evaluation Understudy (BLEU) score of 28.4 on English-to-German translation, surpassing previous results by over 2 BLEU points, and sets a new state-of-the-art BLEU score of 41.0 for English-to-French translation. Moreover, neural machine translation replaces traditional statistical methods with a single neural network for improved translation. Recent models like encoder-decoders encode source sentences into fixed-length

vectors, limiting performance. To address this, [68] propose integrating attention mechanisms, allowing models to focus on relevant parts of the source sentence dynamically.

3.2 Unsupervised learning algorithms

Unsupervised learning involves algorithms that autonomously extract insights from input data without direct guidance or labeled responses. Its objective is to discern latent patterns or structures within the dataset, such as clusters, associations, or distributions, without depending on predefined categories or target outputs [69].

3.2.1 Self organizing maps

The Self Organizing Map (SOM) is an unsupervised ML method that creates a lower-dimensional representation, often in 2D, of high-dimensional data while maintaining its topological structure. SOM finds application in medical image processing for tasks like image segmentation, feature extraction, and data visualization [70].

3.2.2 Autoencoders

AEs are a type of neural networks used in unsupervised learning tasks, particularly for data compression and reconstruction. They consist of an encoder network that compresses the input data into a latent space representation and a decoder network that reconstructs the original input. Autoencoders are trained to minimize the difference between the input and the output, effectively learning to encode and decode data in a compact and meaningful way [71].

3.2.3 Restricted boltzmann machine

Restricted Boltzmann Machine (RBM), an unsupervised generative model, captures the probability distribution of a dataset to infer unseen data, allowing the generation of new samples. It comprises an input layer (visible layer) and one or more hidden layers, forming neural network architecture capable of learning complex representations. The Boltzmann Machine organizes neurons into interconnected layers, both within and across layers, making a unique connectivity pattern that allows the model to grasp intricate dependencies within the data and understand complex relationships [72]. Figure 11 illustrates the architecture of RBM.

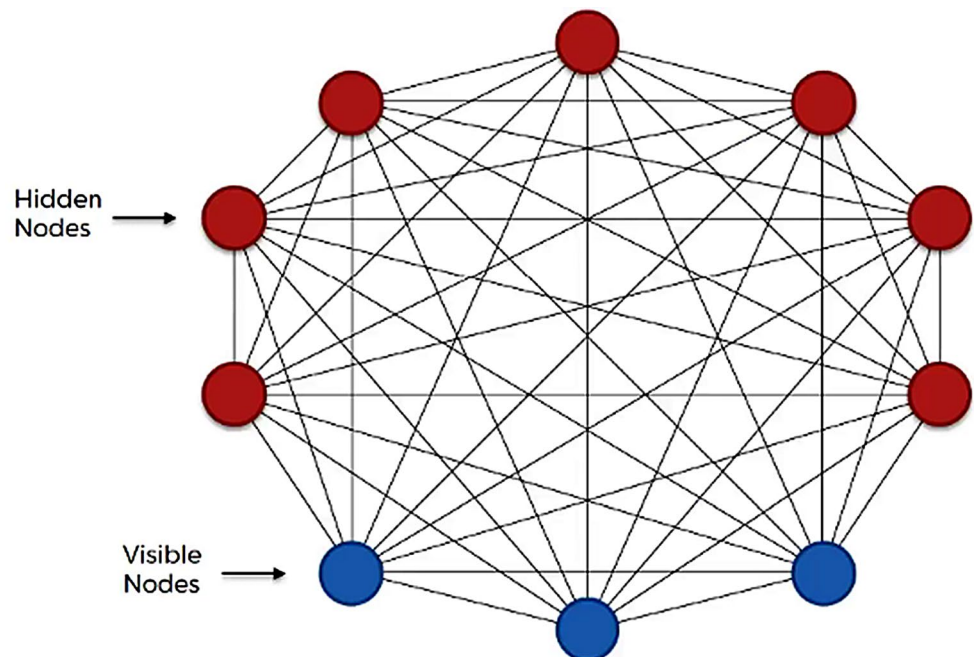
3.2.4 Generative adversarial nets

[73] introduces a novel approach for training generative models using an adversarial process involving two models: a generative model “G” capturing data distribution and a discriminative model “D” estimating sample origin. The system, implemented with multilayer perceptrons, is trained via backpropagation, preventing the need for Markov chains or approximate inference networks [74]. The experimental outcomes demonstrate how well the framework can generate samples.

3.2.5 Deep belief networks

[75] introduces a unique way for training Deep Belief Networks (DBNs), overcoming inference challenges by utilizing “complementary priors.” Their approach involves a fast, greedy algorithm for layer-wise learning and contrastive

Fig. 11 RBM architecture



fine-tuning using a wake-sleep algorithm [76]. The resulting DBN effectively models the joint distribution of handwritten digit images and labels, outperforming discriminative learning algorithms in digit classification tasks. The success of DBNs is attributed to their ability to represent low-dimensional data structures, simplifying exploration and inference tasks.

3.3 Reinforcement learning algorithms

Reinforcement Learning (RL) is a type of learning where an agent learns to make decisions by interacting with an environment. Through trial and error, it discovers the best actions concerning different contexts to maximize cumulative rewards. [77] introduces novel learning techniques for prediction tasks, using experience to predict future system behavior. Unlike traditional methods, these focus on differences between successive predictions rather than comparing predictions to actual results. The techniques are proven effective and efficient for various cases, offering improved accuracy with less memory and computation.

3.3.1 Q-learning

Q-learning is a RL technique by [78] where an agent learns to make decisions by updating Q-values, representing expected rewards for actions in states. It iteratively interacts with the environment, balancing investigation and exploitation. After each action, the Q-values get updated using observed rewards and future predictions. Through this process, the agent refines its decision-making strategy to maximize long-term rewards.

3.3.2 Deep Q-networks

The Deep Q-Networks (DQN) algorithm by [79] introduces a pioneering DL model capable of learning control policies directly from complex sensory input using RL. Utilizing a CNN trained with a Q-learning variant, it processes raw pixel data to estimate future rewards. The model evaluated across seven Atari 2600 games outperforms previous methods or learning approaches.

3.3.3 Proximal policy optimization

Proximal Policy Optimization (PPO) by [80] introduces a new policy gradient method for RL. It alternates between sampling data from interactions with the environment and optimizing a surrogate objective function via stochastic gradient ascent. PPO's novel approach allows multiple mini-batch updates, improving sample efficiency compared to

standard methods. Practical tests show PPO's superior performance and balanced efficiency in various tasks.

3.3.4 State-Action-Reward-State-Action

State-Action-Reward-State-Action (SARSA) operates by estimating the value of actions in different states and updating those estimates based on observed rewards and following states. It follows an iterative exploration process, selecting actions according to its current policy and updating its estimates. Unlike off-policy methods, SARSA is an on-policy algorithm, allowing it to adjust its policy continuously based on real-time experiences. This characteristic makes SARSA well-suited for environments requiring frequent policy modifications to address evolving conditions [81]. In the subsequent section, we explore data leakage issues in the clinical evaluation of classification algorithms and the primary classification areas.

4 Data leakage issues in clinical evaluation of classification algorithms and main classification areas

This section of the paper discusses various data leakage issues associated with DL algorithms for accurate disease diagnosis, followed by an analysis of the leading classification areas in AD diagnosis.

4.1 Data leakage issues and their main causes

An accurate examination of classification algorithms is crucial for ML and DL models, assuring their clinical evaluation remains unbiased. Data leakage, the unintended use of test data during training, introduces bias and undermines the integrity of assessments. Detecting data leakage, especially in DL techniques, poses challenges due to their complexity and adaptability. An extensive examination of data leakage prevalence and causes, detailed in subsequent sections, categorizes data leakage problem into various types like direct expression of data leakage, Indirect or ambiguous manifestation of data leakage etc. The identified causes were subsequently classified based on the specific type of data leakage they entailed, providing further categorization [82]. In this context, four main causes were identified as follows:

4.1.1 Data splitting errors

Inadequate subject-based partitioning of the dataset during the establishment of training, validation, and test sets can result in duplicate data from the same subject or patient across multiple sets, thereby endangering the integrity and

impartiality of the analysis. This issue occurs when patches or slices are selected from a 3D image or when several images of the same subject are accessible at different time points, potentially resulting in data repetition and complicating the analysis. A study by [11] found that biased dataset splitting at the image level in longitudinal datasets can lead to an 8% increase in accuracy compared to a fair data split at the subject level.

4.2 Delayed data partitioning

To maintain fairness in dataset handling, it's crucial to refrain from utilizing the test set for tasks such as data augmentation, feature extraction, and pre-training. These procedures must be conducted after partitioning the training and validation sets. Performing data augmentation after segregating the test data from the training/validation sets helps prevent duplication issues, as it avoids generating images from the same source in both sets [83].

4.3 Transfer learning bias

Transfer learning can introduce a data leakage issue when there is an overlap between the source and target domains. An instance of biased transfer learning occurs when a pre-trained network trained on an AD vs. CN task initializes the network for an MCI vs. CN task. Including the CN patients from the training/validation sets in the source task (AD vs. CN) in the test set of the target task (MCI vs. CN) leads to such a scenario [84].

4.4 Insufficient provision of independent test sets

The evaluation of the classifier's final performance should solely rely on the test set, excluding any utilization of the training hyper-parameters (e.g., learning rate, number of epochs) in the assessment process. To ensure proper hyper-parameter optimization, utilizing a distinct validation set in advance is crucial. Many studies need to pay more attention to the consideration of data leakage during network architecture design, neglecting its potential impact [84].

4.5 Main classification areas

Individuals transitioning towards AD undergo distinct stages, starting with **Cognitive Normalcy (CN)** before progressing to **Mild Cognitive Impairment (MCI)** and eventually **AD**. Differentiating between AD and CN individuals is a critical research focus, along with identifying early AD stages by comparing MCI and CN characteristics. Individuals diagnosed with MCI either **experiences a steady progression** known as **progressive Mild Cognitive Impairment**

(pMCI), leading to **AD**, or **they remain stable, referred to as stable Mild Cognitive Impairment (sMCI)**. Distinguishing AD stages, from CN to MCI and AD diagnosis, are crucial for risk prediction and intervention planning. However, challenges related to data leakage impede the evaluation of classification algorithms, thereby compromising the integrity and objectivity of clinical assessments. This section begins by outlining different data leakage issues, followed by an investigation into AD classification utilizing end-to-end CNNs, 2D slice-level CNNs, 3D patch-level CNNs, ROI-based CNNs, and 3D subject-level CNNs [85].

4.5.1 Classification of AD with end-to-end CNNs

This section examines the utilization of CNNs within an Euclidean domain, focusing on 2D or 3D images within an end-to-end structure from input to classification. The studies are classified based on the network's input type, including 2D slice-level, 3D patch-level, ROI-based, and 3D subject-level inputs. This categorization allows for a comprehensive analysis of the research findings.

4.5.2 2D slice-level CNN

In MRI imaging, the 2D slice-level analysis involves scrutinizing individual slices extracted from the three-dimensional MRI volume. This method allows for a detailed analysis of specific anatomical features or abnormalities by treating each slice as a standalone image. It's commonly applied in tasks such as tumor identification and segmentation within medical image analysis [86]. Many research studies have utilized 2DCNNs with input data comprising a collection of 2D slices extracted from 3D MRI volumes [87]. This approach offers the advantage of leveraging the effectiveness of well-established CNNs, which have demonstrated remarkable success in natural image classification. These pre-existing CNN models can be harnessed in the transfer learning process, facilitating improved performance and efficiency. Moreover, from a single 3D image, it is possible to extract multiple distinct slices, thereby increasing the number of available training examples. A study by [88] compares five models—ResNet, Convolutional NeXt(ConvNeXt) [89], Class-Attention in Image Transformers (CaiT) [90], Swin Transformer [91], and Convolution Vision Transformer (CVT) [92]—on AD detection using 2D image slices. ConvNeXt demonstrates superior performance, achieving 95.74% accuracy. These findings underscore the potential of ConvNeXt in AD identification via 2D slice analysis. 2D slice-level CNNs efficiently examine individual brain slices in diagnosing AD, improving lesion detection accuracy. However, they might require additional support in understanding the spatial connections

between slices, which could decrease diagnostic accuracy, especially when dealing with volumetric brain imaging. A significant limitation of the 2D slice-level approach is its dependence on independently analyzing the original three-dimensional MRI data per slice using a 2D convolutional layer. Also, selecting slices for input involves various techniques, introducing potential ambiguity in distinguishing between slice-level and subject-level accuracy.

4.5.3 3D patch-level CNN

Researchers are exploring an innovative approach, termed the 3D patch-level CNN technique, to overcome the limitations of the 2D slice-level CNN approach. The 3D patch-level approach involves using 3D patches as input, reducing memory usage, which is beneficial in resource-limited scenarios. Its advantage in AD diagnosis stems from its ability to incorporate 3D information from MRI images, effectively capturing spatial relationships within the brain compared to individual 2D slices, which may improve classification accuracy. Maintaining consistency in neural network architecture and parameters across all patches is vital for optimizing processing. While this uniformity is crucial, certain research studies have explored utilizing larger patches as an alternative strategy [13, 14]. An active learning-based sampling strategy to address the scarcity of medical data, concentrating on complex brain imaging structures, is introduced by [93]. Their study utilized a patch-level approach alongside a multi-instance learning framework, improving information extraction while reducing data acquisition and annotation costs. Validation of the ADNI dataset confirms the approach's effectiveness. [20] tested patch sizes ($15 \times 15 \times 15$ to $45 \times 45 \times 45$) for their proposed H-FCN method, maintaining a fixed input patch size of $(25 \times 25 \times 25)$. Optimal performance is achieved with patch sizes between $(25 \times 25 \times 25)$ and $(35 \times 35 \times 35)$, while smaller patches ($15 \times 15 \times 15$) lack global structural information. In their study, larger patches ($45 \times 45 \times 45$) showed reduced performance, likely due to including more uninformative voxels. 3D patch-level CNNs are good at capturing spatial relationships in volumetric medical images, ensuring accurate feature representation. However, they require more computational power and take longer to train than 2D methods, which can be challenging, particularly in settings with limited resources. In a study [94], separate CNNs are pre-trained for each patch, with their outputs merged to diagnose a subject. Conversely, [95] has used a different approach, where patches are not directly associated with diagnoses. Therefore, they train the ensemble network from the beginning rather than training individual CNNs per patch before ensemble learning. [96] used smaller patches ($32 \times 32 \times 32$), necessitating consideration of potential discrepancies

between patches taken at identical coordinates for different subjects, which they addressed to ensure consistency. In medical imaging, 3D patch-level CNNs are good at capturing spatial relationships for precise feature extraction and improved diagnostic accuracy. However, compared to ROI-based CNNs, they may demand more considerable computational resources and longer training times due to processing volumetric data rather than focusing solely on specific regions.

4.5.4 ROI-based CNN

3D patch-level techniques divide MRIs into smaller inputs, but some patches may lack disease-related content. ROI-based approaches target key brain areas, simplifying the framework by reducing neural network inputs. The hippocampus, sensitive to early AD effects, is commonly studied. Professionals analyze AD prediction accuracy using individual ROIs and try various combinations, rigorously training and validating models, and then evaluate performance metrics like accuracy, specificity, etc. [97]. Some studies have conducted linear registration [98] and delineated a 3D bounding box including all hippocampal voxels through segmentation utilizing the Automated Anatomical Labeling (AAL) atlas [99]. These three studies adopted a “2D + ϵ approach,” where patches comprised three neighboring 2D slices within the hippocampus. The “2D + ϵ approach” involves generating patches by combining 2D slices, typically with a slight overlap (ϵ) between neighboring slices. This technique leverages the benefits of 2D analysis while incorporating contextual information from adjacent slices, allowing more comprehensive information extraction from volumetric data like medical images. They utilize one or three patches per patient, which may not confine the entire region. [9] exclusively utilizes the sagittal view and assigns one patch per patient for classification. The CNN architecture incorporates two convolutional layers with associated max pooling and one fully connected layer. In [12], patches are generated using all views (sagittal, coronal, and axial), with three patches being created per subject. Three networks are then trained for each view and, after that, fused. [10] focuses on transferring learning from anatomical MRI to diffusion MRI, which involves advanced concepts. In [100], non-linear registration [101] is conducted to establish voxel correspondence between subjects, followed by the identification of hippocampal voxels through segmentation using Multi-Atlas Label Propagation with Expectation-Maximization (MALP-EM) [102]. Per image, one hundred fifty-one patches were used, with fixed sampling positions maintained throughout the experiments. Each of these consists of the concatenation of three 2D slices along the three possible planes (sagittal, coronal, and axial), originating

from one voxel belonging to the hippocampus. One significant drawback of this approach is its limited analysis, focusing solely on one or a few regions despite the widespread distribution of AD-related changes across multiple brain areas. However, it effectively tackles the challenge of overfitting by utilizing a reduced number of inputs, in contrast to methods that integrate patch combinations.

4.5.5 3D subject-level CNN

With the rise in high-performance computing resources in recent years, more studies have adopted a 3D subject-level approach for medical imaging analysis. This method utilizes the entire MRI scan and performs classification at the subject level. The essential benefit lies in the comprehensive integration of spatial information, as the CNN processes the whole 3D volume without needing patch-wise processing. Several research efforts have adapted classical architectures, ResNet [103] and VGGNet [104], to effectively process entire MRI scans [19]. In both instances, the classification accuracies from VGGNet and ResNet are similar, but they fall short of the highest accuracies achieved by other 3D subject-level approaches. [105] experimented with complex modules inspired by classical architectures of pre-trained models incorporating elements like dilated convolutions, dense blocks, and residual blocks, using the 3D subject-level CNN approach. [106] utilizes 3D subject-level CNNs to detect AD using brain images of regular CN, Early Mild Cognitive Impairment (EMCI), and AD patients. The model achieves impressive accuracy rates of 96% in three-way classification and 97% in distinguishing between NC and Alzheimer patients. Moreover, the Gradient Weighted-Class Activation Mapping (Grad-CAM) [107] visualization indicates the model's focus on brain ventricles and the hippocampus, providing insights into AD progression. Table 1 outlines studies utilizing various CNN strategies for AD classification using anatomical MRI data. Part

(A) of the table lists studies without reported data leakage including [9], [10], [11], [19] [20], [21], [94], [95], [96], [105], [108], [109], while Part (B) includes studies where potential data leakage was noted or suspected. The studies [8], [11], [15], [25], [28], [100], [110], [111], [112], [113], [114], [115], [116] and [117] have potential data leakage concerns. These arise from errors in data splitting, insufficient provision of independent test sets, delayed data partitioning, and transfer learning bias. A specific study by [11] introduced deliberate data leakage to evaluate its impact, simulating scenarios across both investigated categories (See Tables 1 and 2).

In the next section, the paper discusses the role of transfer learning in disease diagnosis tasks.

5 Role of transfer learning in disease diagnosis

Transfer learning is a method in DL that has transformed how neural networks are constructed and trained. It introduces innovative techniques to improve network development and training, leveraging pre-trained models to transfer knowledge seamlessly between tasks, driving significant progress across various domains. Transfer learning addresses small dataset limitations by leveraging extensive data and complex models to improve accuracy, efficiency, and generalization. This approach expands horizons by connecting different domains and facilitates the transfer of learned features. It reduces the necessity for extensive task-specific training and lowers computational demands, making the DL process more efficient. Figure 12 presents a visual representation of the fundamental idea behind transfer learning.

Pretrained models such as VGG [104], ResNet [103], Inception [118], MobileNet [119], DenseNet [120], EfficientNet [121] etc., trained initially on ImageNet [122], are

Table 1 None detected

Authors	Performance Accuracy					Strategy
	AD vs. CN	sMCI vs. pMCI	MCI vs. CN	AD vs. MCI	Multi-class	
[9]	Ac = 82.8%		Ac = 66%	Ac = 62.5%*		ROI-based
[108]	Ac = 81%				Ac = 57% ^{II}	2D slice-level
[19]	Ac = 80%					3D subject-level
[21]	Ac = 88%					3D subject-level
[10]	BAc = 90%		BAc = 73%	BAc = 83%		ROI-based
[11]	Ac = 90%					3D subject-level
[109]					Ac = 93% ^I *	2D slice-level
[96]	Ac = 90%		Ac = 74%*			3D patch-level
[20]	Ac = 90%	Ac = 80%*				3D patch-level
[95]	Ac = 91%	Ac = 78%*				3D patch-level
[94]	Ac = 91%					3D patch-level
[105]	Ac = 76%		Ac = 75%	Ac = 76%		3D subject-level

Table 2 Data leakage table

Reference	Performance Accuracy					Strategy	Data leakage (type)
	AD vs. CN	sMCI vs. pMCI	MCI vs. CN	AD vs. MCI	Multi-class		
[110]	Ac=96%					2D slice-level	Wrong split
[8]	Ac=91%		Ac=66%	Ac=70%		ROI-based	Unclear (ii, iii)
[25]	Ac=85%					3D subject-level	Unclear (i)
[111]					Ac=74% ^I *	2D slice-level	Unclear (ii, iii)
[15]					Ac=99% ^{III} *	2D slice-level	Wrong split
[112]			Ac=90.6%			2D slice-level	Wrong split and augmentation before split
[113]					Ac=96% ^{II}	3D subject-level	Wrong split
[114]	Ac=99%		Ac=94%	Ac=100%	Ac=95% ^{II}	3D subject-level	Unclear (i)
[100]	Ac=89%	Ac=73%				ROI-based	Unclear (ii)
[115]	Ac=100%					2D slice-level	Unclear (ii)
[116]	Ac=86%		Ac=86%	Ac=77%	Ac=80% ^{II}	3D subject-level	Clear (i, iii)
[117]	Ac=85%	Ac=74%				3D patch-level	Unclear (iv)
[11]	Ac=99%					3D subject-level	Clear (i)
[28]					Ac=95% ^{IV} *	2D slice-level	Clear (i, ii)

“* This metric reflects the accuracy achieved on a highly imbalanced dataset, where one class constitutes less than half of the other. This imbalance can result in an overly optimistic estimation of performance.” Other terminologies used are:-

^ICN vs. mild vs. moderate vs. severe,

^{II}AD vs. MCI vs. CN,

^{III}AD vs. LMCI vs. EMCI vs. CN,

^{IV}sMCI vs. pMCI vs. CN

Ac: Accuracy

BAc: Balanced Accuracy

extensively utilized in transfer learning. These models are versatile tools that can extract data features or be fine-tuned for specific tasks. By leveraging these pre-trained models, researchers can significantly reduce the required training and computational resources needed for DL tasks. Training the fine-tuned model on a dataset consisting of neuroimaging biomarkers like MRI or PET scans, in conjunction with corresponding stage labels, allows for the extraction of relevant features and enables accurate classification. One of the study by [123] used pre-trained models like AD-VGG16, AD-Resnet50, and AD-2DCNN were concurrently used to improve diagnostic accuracy for AD using MRI images. The study revealed varied performance levels among the models, with AD-2DCNN demonstrating the highest accuracy of 98.28% on the test set. This comparative evaluation underscores the utility of pre-trained DL models in AD diagnosis.

One of the studies by [124] aimed to predict AD using socio-demographic, clinical, and MRI data from OASIS-1. They utilized a Gradient-Boosted Machine (GBM) [125] to predict AD presence and severity, achieving 91.3% accuracy. By incorporating ResNet-50, their study has achieved

98.99% accuracy in identifying AD patients from MRI images, showcasing its potential for automated AD diagnosis. In [126], researchers utilize transfer learning with pre-trained DL models like GoogLeNet [118], VGG-16, AlexNet [127], and ResNet-18 to classify MR images into AD, MCI, and NC. GoogLeNet achieves the highest accuracy of 99.84% for training and 98.25% for testing, surpassing other models. In their study, [23] conducted a comprehensive investigation into the cutting-edge advancements in early AD detection. They utilized DL models in conjunction with transfer learning and neuroimaging biomarkers to achieve their research objectives. Thorough screening resulted in the identification of many studies that met the inclusion criteria and all have used transfer learning approach. The findings demonstrate that the utilization of transfer learning supports researchers in creating a more precise system for the early detection of AD.

[128] proposes leveraging transfer learning and tissue segmentation in MRI-based detection of early AD. Utilizing a step-by-step transfer learning method with a pre-trained VGG architecture, the model effectively distinguishes between NC, EMCI, LMCI, and AD patients. The proposed

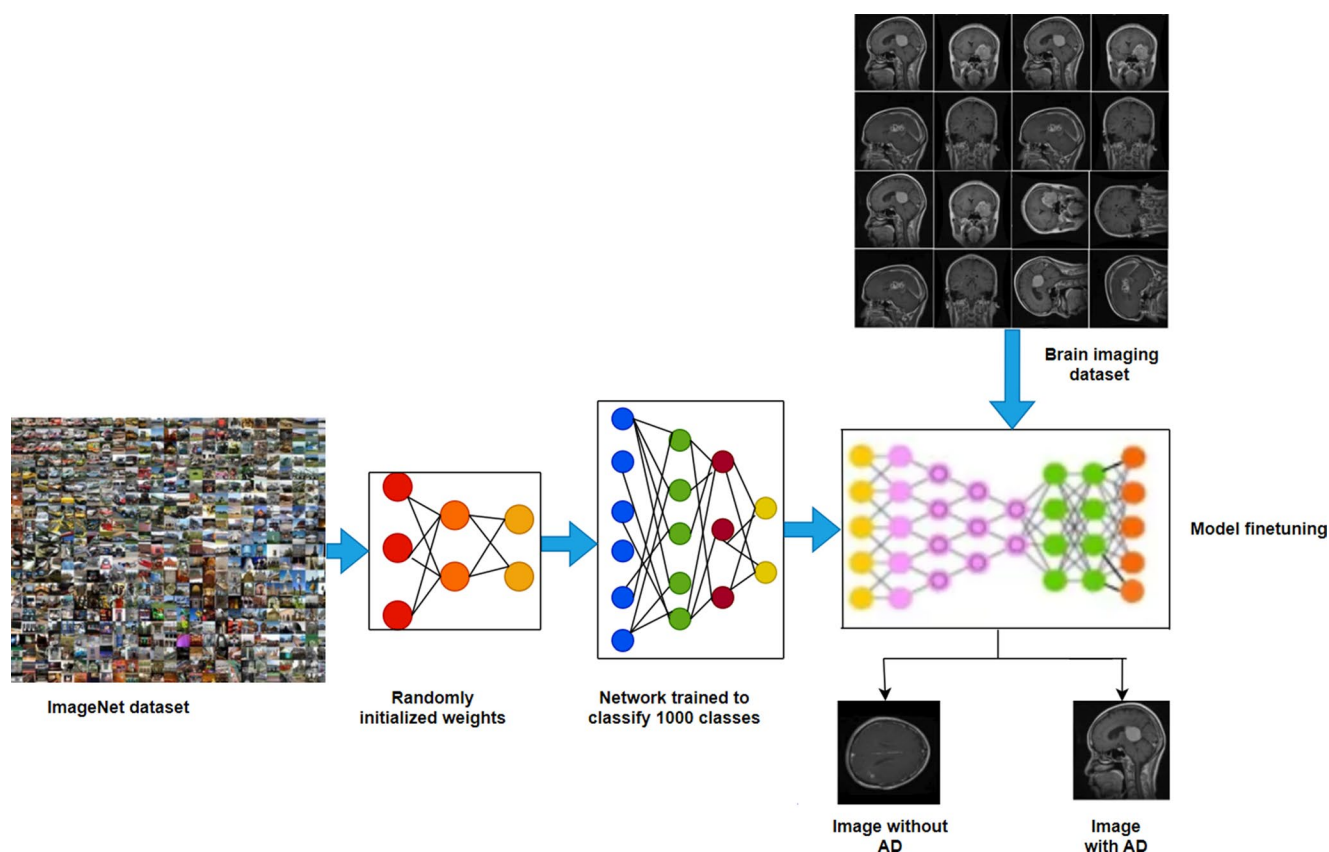


Fig. 12 Idea behind transfer learning

approach attained remarkable classification accuracies by integrating tissue segmentation and testing on the ADNI database, achieving 98.73% for distinguishing AD from NC and 83.72% for discerning EMCI and LMCI. These results highlight the efficacy of transfer learning in advancing AD diagnosis, as evidenced by comparative evaluations against established models. [129] introduces a DL framework for AD classification, emphasizing transfer learning with AlexNet and LSTM. When integrated with transfer learning and addressing overfitting using a GAN for data augmentation, their framework achieves high accuracies: 98.13% for AD-CN, 99.38% for AD-MCI, and 99.37% for CN-MCI. The model demonstrates superior performance when trained on the ADNI dataset by achieving a 96.83% accuracy in multiclass classification and outperforming other techniques in accuracy, sensitivity, and hypothesis testing. These findings underscore the significant contribution of deep transfer learning in progressing brain tumor diagnosis, providing valuable insights for forthcoming medical imaging endeavors. The subsequent section discusses various imaging modalities and neural network scans for the detection of AD.

6 Various imaging modalities / neural network scans for Alzheimer's disease detection

Different imaging methods utilized in AD detection via DL include Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET), Computer Tomography (CT), Single-Photon Emission Computed Tomography (SPECT), Functional Magnetic Resonance Imaging (fMRI), and Diffusion Tensor Imaging (DTI).

6.1 Alzheimer's disease detection using deep learning via structural MRI dataset

Structural MRI is a valuable tool for AD detection, allowing for examining brain structures affected by the disease. It enables the identification of specific changes associated with Alzheimer's, such as hippocampal atrophy and cortical thinning, aiding in early diagnosis and monitoring disease progression. Structural MRI plays a crucial role in research and clinical practice related to AD by providing detailed images of the brain's structure. MRI imaging provides high-resolution images of soft tissues and organs without radiation exposure, offering multiplanar capabilities for

comprehensive anatomical views, and is versatile in diagnosing medical conditions. However, drawbacks include high costs, long scan times, contraindications for some individuals, and limited machine availability [130]. Structural MRIs were used by [131] for AD detection via DL by proposing a model-based Siamese Convolutional Neural Network (SCNN) inspired by VGG-16 architecture. The approach used in this study involves utilizing a deep triplet network to analyze brain MRI data and detect AD. This network incorporates a conditional loss function to address the challenge of limited sample availability and improve model accuracy. The model, trained on the OASIS dataset with MRI scans across dementia stages, achieves a remarkable 99.41% accuracy in predicting multi-class dementia stages.

6.2 Alzheimer's disease detection using deep learning via functional magnetic resonance imaging dataset

fMRI imaging provides benefits in non-invasively mapping brain activity, helping research and clinical diagnosis by providing insights into cognitive functions, thereby assisting in pre-surgical planning or disease diagnosis. However, it faces limitations, including its modest temporal resolution that may lead to inaccuracies in capturing rapid neural activity and its restricted ability to identify specific neural circuits precisely. Furthermore, the subjective analysis of fMRI data might affect the reliability and reproducibility of results, highlighting the importance of conscientiously addressing factors like participant cooperation and experimental design [132]. A study by [133] used fMRI images to detect AD by analyzing the relationship between the fMRI images of Alzheimer's patients and their scores on the Mini-Mental State Examination (MMSE). The MMSE score categorizes Alzheimer's severity into low, mild, moderate, and severe. In their study, the extracted features are used with ML models like K-Nearest Neighbors (KNN), SVM, decision tree, linear discriminator analysis, random forest, and CNN to accurately classify and diagnose Alzheimer's stages, with the presented CNN architecture achieving superior performance.

6.3 Alzheimer's disease detection using deep learning via fluorodeoxyglucose positron emission tomography dataset

AD detection using DL on FDG-PET datasets shows promising results in accurately identifying patterns of brain metabolism associated with the disease. By leveraging the power of DL algorithms, PET-based approaches offer potential for improved diagnostic accuracy and early detection of AD. PET is commonly used to measure glucose levels in

the brain, indicating brain metabolism and synaptic activity due to the brain's reliance on glucose for energy. Radiologists diagnose AD by detecting the positrons emitted when radioactive-tagged glucose is consumed by the brain, indicating lower glucose levels across the brain. FDG-PET stands out for its remarkable sensitivity in pinpointing metabolic irregularities. It allows comprehensive body scans and detailed analysis of glucose metabolism, which aids in early disease identification and tracking treatment progress. However, it faces challenges such as its restricted ability to precisely locate lesions, radiation exposure concerns, potential for false-positive results, high financial burden, and limited accessibility [134]. The study [135] utilized FDG-PET data to evaluate the reduction in the regional cerebral metabolic rate for glucose, which is a reliable metabolic biomarker for AD. Their work introduced an innovative approach utilizing DL techniques to accurately differentiate between different diagnostic categories of Alzheimer's using FDG-PET data. Using the ADNI2 dataset, the study validates the proposed approach for AD classification with FDG-PET data, achieving competitive results in classification accuracy.

6.4 Alzheimer's disease detection using deep learning via diffusion tensor imaging dataset

DTI datasets offer significant advantages in diagnosing various diseases by providing detailed insights into tissue microstructure and integrity. Specifically notable in neurology, DTI enables quantitative analysis and tractography, improving understanding of tissue organization and pathology across medical fields. However, challenges such as susceptibility to motion artifacts, limited spatial resolution, complex post-processing requirements, and sensitivity to imaging parameters exist, demanding careful consideration during data interpretation and clinical use. However, despite these challenges, DTI remains highly valuable in revealing tissue pathology and improving diagnostic capabilities across various diseases [136]. A shallow architecture network is used to develop a robust classification system for AD and MCI against NC [137]. The results underscore data integration using techniques from various imaging modalities, such as structural MRI and DTI. Their study also proposes a method to confirm system robustness by incorporating multiple scans per subject, opening avenues for improving AD treatment strategies.

6.5 Alzheimer's disease detection using deep learning via computed tomography dataset

CT datasets are essential for disease diagnosis, providing quick and detailed body imaging with versatile capabilities across anatomical regions. They aid in detecting conditions

like tumors, fractures, and internal bleeding, enhanced by contrast agents. Although CT scans are readily accessible, they present concerns regarding radiation exposure, contrast allergies, limitations in soft tissue contrast, and susceptibility to artifacts [138]. A study by [139] proposed a system that achieved an impressive accuracy rate of 98.32% in accurately identifying AD using neuroimaging data. Their study devised VGG-16 & Improved Faster Recurrent Convolutional Neural Network (IFRCNN) using the ImageNet database. This technique utilizes a mathematical model originally crafted for action recognition to extract features for categorizing tasks. The analysis's results underscore the efficacy of ML for classifying AD using CT images. After, exploring various imaging modalities and neural network scans for detecting AD, the subsequent section delves into datasets and databases related to AD.

7 Datasets and databases related to Alzheimer's disease

This section initially explores the fundamental aspects of brain imaging formats and preprocessing requirements, which are crucial for thoroughly understanding brain imaging techniques and data preprocessing methods. The discussion then explores the datasets and databases relevant to AD, emphasizing their importance in research and clinical settings.

7.1 Brain imaging formats and preprocessing requirements

Brain imaging techniques, non-invasive in nature, help examine brain structure, function, and medication effects without invasive procedures. These methods include structural imaging, revealing brain composition, and functional imaging, elucidating brain activity [140]. fMRI data, acquired in various formats, commonly utilizes Digital Imaging and Communications in Medicine (DICOM), a universal standard for medical images [141]. DICOM consolidates image slices and metadata into single files, easing storage and sharing while preserving vital header information for specialized software extraction. An initiative has been launched to establish a novel 4D imaging format called the Neuro Imaging Informatics Technology Initiative (NIFTI) [142], supported by the National Institutes of Health. The NIFTI team works to enhance the compatibility and smooth integration of functional neuroimaging software products. Data in NIFTI is stored in either a single or dual file format. In the "dual file" format of NIFTI, the data is stored in a pair of files: an information-containing header file (.hdr) and an image data file (.img). In the "single file"

format, the data is stored in a .nii file, where the header information precedes the image data. Recent advancements in Python [143] have resulted in the creation of specialized packages designed for loading NIFTI data. One such package is `dicom2nifti`, developed to convert DICOM images into NIFTI format [144]. Another is `NiBabel`, which facilitates the reading and conversion of NIFTI files into various neuroimaging formats. Further, `NiLearn` empowers statistical analysis and ML on neuroimaging data by providing utilities for reading, writing, manipulating, and visualizing NIFTI images [145].

Pre-processing is a vital methodology in data mining, converting raw data into an understandable and analytically viable format. This crucial step filters out irrelevant elements and adjusts the data for better comprehension and analysis. In the domain of fMRI data analysis, pre-processing holds particular significance as it confirms the removal of unwanted abnormalities and standardizes the data format. Commonly utilized tools like Statistical Parametric Mapping (SPM) [146], Analysis of Functional Neuro Images (AFNI) [147], Functional Magnetic Resonance Imaging of the Brain (FMRIB) Software Library [148], and FreeSurfer [149] play essential roles in neuroimaging data processing. Pre-processing methods are pivotal for eliminating extraneous non-brain tissues, including the skull region, and enhancing the clarity of brain imaging. Emphasizing the significance of pre-processing and feature extraction techniques in brain image categorization and segmentation, MRI scans typically appear as 3D NIFTI volumes, capturing intricate imaging details. The commonly utilized SPM-8 tool assists in tasks such as skull stripping and gray matter segmentation by utilizing spatial normalization, bias correction, and modulation techniques. The Python `NiBabel` packages convert gray matter volumes into individual JPEG slices. During this process, the initial and final slices of the dataset, which hold minimal information, are omitted. Despite the remarkable capability of CNNs in extracting features from raw images, image preprocessing remains paramount in the overall workflow. Once the registration concludes, each image undergoes thorough visual quality inspections to ensure its integrity and overall appearance [150].

7.2 Dataset and database

In recent decades, many innovative neuroimaging techniques have emerged, leading to the development of vast multi-modal neuroimaging datasets. Consequently, neuroimaging researchers started progressively releasing more datasets to the public, yielding valuable results. As an outcome, there has been a notable rise in the accessibility of datasets, mainly targeting brain diseases such as AD. A variety of datasets and databases are available for detecting

AD, including well-known ones like the “Alzheimer’s Neuroimaging Initiative (ADNI),” the “Australian Imaging Biomarkers and Lifestyle (AIBL),” “Open Access Series of Imaging Studies (OASIS)” etc. A brief description of these datasets and databases, explaining the associated diagnosis labels, is provided. It’s essential to recognize that the diagnostic criteria may differ among these studies, meaning the labels assigned to ADNI, AIBL, OASIS, and other datasets vary slightly.

7.2.1 Alzheimer’s disease neuroimaging initiative

The ADNI dataset, derived from the ADNI database (www.adni.loni.usc.edu), is among the most extensively utilized datasets for detecting AD. In 2004, the ADNI was established as a collaborative effort between public and private commodities led by Principal Investigator Dr. Michael W. Weiner. The primary objective of ADNI is to evaluate the potential of sequential **MRI, PET, clinical/neuropsychological assessments, and other biological markers** in monitoring the progression of MCI and AD [151]. For the most up-to-date information, interested readers can refer to www.adni-info.org. The ADNI study comprises four distinct cohorts: **ADNI-1, ADNI-GO, ADNI-2, and ADNI-3**. Given these cohorts’ interdependence and longitudinal nature, the same individual can be present in multiple cohorts and across different cohorts over time [152].

The study categorizes participants into distinct groups based on their cognitive health status. These groups are referred to as CN (Cognitively Normal), MCI (Mild Cognitive Impairment), sMCI (Stable MCI), pMCI (Progressive MCI), and AD (Alzheimer’s Disease). CN individuals refer to cognitively normal people who maintain this status throughout the follow-up diagnostic sessions. MCI individuals, whether classified as early MCI (EMCI) or late MCI (LMCI), did not show frequent reversals or transitions and did not revert to a CN state during their diagnostic sessions. Participants classified with sMCI (Stable MCI) received an initial diagnosis of MCI, either as early MCI (EMCI) or late MCI (LMCI), and did not develop AD within 36 months from their current cognitive state. Subjects with pMCI (Progressive MCI) are those who were initially diagnosed with MCI, including EMCI or LMCI, but subsequently developed AD within 36 months after the current session. Individuals with AD state depict those who were diagnosed with AD during the initial assessment.

7.2.2 Australian imaging biomarkers and lifestyle

Similar to the ADNI study, the AIBL dataset is being widely used by researchers for AD detection. AIBL aims to identify **biomarkers, cognitive traits, and lifestyle factors**

contributing to early AD detection [153]. The **dataset comprises longitudinal cohort participants**, with suspected AD individuals undergoing examinations at various intervals to monitor disease progression. Besides clinical and imaging data (MRI and PET), blood and Cerebrospinal Fluid (CSF) sample analysis is crucial for early AD identification. For more details about accessing this dataset, researchers can visit www.aibl.csiro.au.

AIBL helps in understanding AD across its three stages: preclinical, prodromal, and clinical, while also shedding light on the disease’s biological progression. The early use of advanced brain imaging scans and improved blood tests helps researchers explore how genetic and environmental factors affect when AD starts and how it progresses. The AIBL dataset constitutes a significant and extensively documented resource, containing subjects with comprehensive longitudinal cognitive, imaging, CSF sample collections, and lifestyle data. This dataset is invaluable for evaluating interventions to delay, inhibit, or treat AD.

7.2.3 Open access series of imaging studies

The dataset consists of three separate cohorts, namely **OASIS-1, OASIS-2, and OASIS-3**, which can be accessed via the website <http://www.oasis-brains.org/>. The **initial cohort exclusively contains cross-sectional data, while the two subsequent cohorts include longitudinal data**. Different diagnostic labels are utilized, taking into account the Clinical Dementia Rating (CDR) scale. The OASIS dataset comprises three distinct cohorts: OASIS-1, OASIS-2, and OASIS-3, accessible through the website <http://www.oasis-brains.org/>. **OASIS-1 features solely cross-sectional data, while OASIS-2 and OASIS-3 include longitudinal data**. Various diagnostic labels are used, incorporating the CDR scale [154].

The **OASIS-1 dataset comprises two distinct labels: the AD category and the control category**. The AD category encompasses subjects with a non-null CDR score, indicating progression beyond MCI but not yet reaching the severity of AD as classified in ADNI. Subcategories within this group are delineated based on the 9-point CDR scale, where scores of 0.5, 1, 2, and 3 correspond to very mild, mild, moderate, and severe dementia, respectively. The ADNI and AIBL datasets follow a standardized method where healthcare experts categorize individuals into three diagnostic groups (CN, MCI, and AD) based on thorough clinical evaluations, as detailed in robust research studies [152, 153]. The labeling systems utilized in ADNI, AIBL, and OASIS lack direct correspondence or strict equivalence due to variations in the diagnostic criteria used across these datasets. The OASIS initiative aims to provide freely accessible neuroimaging datasets for research purposes, contributing

to advancements in neuroscience akin to initiatives such as ADNI. The OASIS-3 dataset presents an expansive accumulation of data spanning 30 years and involving data from over 1000 subjects. It includes over 2000 MRI sessions, primarily consisting of volumetric segmentation files produced using Freesurfer [150].

7.2.4 The national Alzheimer's coordinating center database

The National Alzheimer's Coordinating Center (NACC) dataset oversees and preserves patient data collected from Alzheimer's Disease Centers (ADCs) funded by the National Institute on Aging (NIA). Each center gathers specific participant information and shares it with the NACC minimal dataset. Data management practices vary across centers to meet diverse research needs. Some centers use individual computers with spreadsheet software, while others use networks of servers running sophisticated data management systems like Oracle. NACC faces the challenge of consolidating and structuring previously collected data into a cohesive database that effectively meets administrative and research objectives. NACC allows centers to contribute data according to their specific needs by establishing a database with multiple distinct datasets designed for this purpose. The ADCs clinical task group, designated by the NIA, supervises the Uniform Data Set (UDS), which aims to standardize evaluation practices over time for individuals with mild AD and MCI. The NACC website (www.alz.washington.edu) facilitates data access, with the ongoing expansion of the NACC database combining data components into a fully integrated database crucial for scientists in ADCs investigating various aspects of AD [155].

7.2.5 NIA genetics of Alzheimer's disease data storage site

NIAGADS operates as a national repository housing genomic data, offering authorized researchers access to genotypic and sequencing data for studying AD and related neurological conditions. Collaborating with esteemed consortia and centers such as the Alzheimer's Disease Genetics Consortium (ADGC), Cohorts for Heart and Aging Research in Genomic Epidemiology (CHARGE) Consortium, the Alzheimer's Disease Sequencing Project (ADSP), and the Genome Center for Alzheimer's Disease (GCAD), NIAGADS effectively manages extensive AD datasets. These datasets are readily public and broadly utilized by the research community, solidifying NIAGADS' prominent role in the field. The genomics database hosted by NIAGADS offers a user-friendly interface that facilitates the detection and identification of genes, Single-Nucleotide Polymorphisms (SNPs), and significant genomic regions linked to

AD. It provides researchers with a comprehensive toolkit for effectively utilizing their research findings, encompassing various features like browsing, bookmarking, searching, saving, and sharing results [156].

NIAGADS effectively supports AD researchers in meeting their research goals by encompassing datasets from multiple projects and institutions, streamlining the process, and efficiently fulfilling their requirements. Datasets, guidelines, and newly introduced features are accessible through the website www.niagads.org. NIAGADS is supported by a cooperative agreement with the NIA. Mandatory guidelines dictate that all data derived from NIA-funded studies on AD genetics must be submitted to NIAGADS or another repository approved by the NIA. NIAGADS manages the Data Sharing Service (DSS), allowing authorized users in neurodegenerative research to access and exchange genomic data and findings easily. This simplified process improves accessibility and promotes collaboration among investigators [157].

7.2.6 The AD knowledge portal: a repository for Multi-omic Data on Alzheimer's disease and aging

The AD Knowledge Portal is a centralized repository that houses a vast collection of multi-omic data related to AD and aging. It offers researchers access to diverse data types, including genomics, transcriptomics, proteomics, metabolomics, and imaging data. The portal facilitates data exploration, pattern identification, and potential therapeutic target discovery. The AD Knowledge Portal offers an easy-to-use interface and search features for simplified data exploration. Its analysis and visualization tools allow researchers to extract valuable insights. By fostering collaboration and information exchange, the portal speeds up scientific progress in AD and aging research, improving comprehension of these disorders by facilitating the creation of improved diagnostic and treatment approaches. Researchers generate, scrutinize, and evaluate vast, concise data from diverse human samples and experimental models across various analytical levels. The primary aim of developing the website was to facilitate unrestricted examination and broader utilization of resources produced within the NIA's translational research portfolio. The portal was established under the Accelerating Medicines Partnership in Alzheimer's Disease (AMP-AD) program, offering additional information and exploration avenues at the following link: <https://www.nia.nih.gov/research/amp-ad>. Launched in 2014, the AMP-AD program represents a joint endeavor by the NIH, various philanthropic organizations, and pharmaceutical companies working together towards a common objective. The portal's scope has expanded to support four additional research programs investigating different aspects of aging and AD [158].

7.2.7 LORIS: a web-based data management system for multi-center studies

LORIS (Longitudinal Online Research and Imaging System) is an adaptable internet-based dataset that encompasses every aspect of a multi-center study, encompassing a wide range of data acquisition sources, including imaging, clinical, behavioral, and genetics. Additionally, it comprises data processing, storage, and the eventual dissemination of data. Initially developed to support MRI research on typical brain development funded by the NIH, LORIS was established as a platform to improve the administration of studies, centralize data storage, and allow seamless retrieval of data from various research sites. It offers a structured and systematic framework that eliminates the need to spend excessive time managing extensive variables across multiple collaborators, assuring efficiency and effectiveness in research collaboration. Researchers can securely store and retrieve various data types, including cross-sectional and longitudinal data, effectively facilitating data extraction when needed. The LORIS dataset provides a significant advantage by integrating various data types and modalities, including imaging data, cognitive and behavioral assessments, and chromosomal biomarkers. This comprehensive integration allows for a complete approach to data analysis. LORIS is widely acknowledged as the leading dataset for clinical research, known for its comprehensiveness and user-friendly interface. To learn more, interested researchers can visit www.cbrain.mcgill.ca/loris. Individuals can access the system through a standard web browser, allowing them to perform tasks such as 3D image visualization, data entry, and quality assessment, thereby expanding their range of capabilities [159].

7.2.8 The minimal interval resonance imaging in Alzheimer's disease

The Minimal Interval Resonance Imaging in Alzheimer's Disease (MIRIAD) dataset comprises longitudinal volumetric T1 MRI scans from 46 individuals diagnosed with mild to moderate AD and 23 control subjects, offering valuable research data. The study assured standardized and longitudinal data collection by utilizing a consistent radiographer and scanner who conducted 708 scans at predetermined intervals, including 2, 6, 14, 26, 38, and 52 weeks and 18 and 24 months post-baseline scan. During the scan analysis, factors such as gender, age, and MMSE scores were considered, establishing further correlations between gender, age, and cognitive assessment scores. The analysis focused on examining the gender, age, and MMSE scores of each participant, particularly highlighting the associations between gender, age, and cognitive assessment scores. The main goal of the

MIRIAD project is to determine the optimal time frame for conducting clinical trials on AD by utilizing atrophy measurements from successive MRI scans as a crucial metric. This involves assessing disease progression and evaluating potential therapeutic interventions. The scanning schedule implemented in the project intentionally includes a diverse range of intervals between scans, ranging from 2 weeks to 2 years. This approach allows for a comprehensive collection of data points, aiding in understanding disease progression over various timeframes. Longitudinal back-to-back scanning ensures imaging availability at different time points, enabling sequential data acquisition over multiple periods for analyzing changes and progression. The data is accessible as a shared resource through <http://miriad.drc.ion.ucl.ac.uk/>, allowing the researchers to utilize, evaluate, and compare methodologies for measuring longitudinal volume changes in serially acquired MRIs. This accessibility promotes exploring and advancing techniques for quantifying and analyzing structural changes over time in AD research [160].

7.2.9 Critical path institute online data repository: coalition against major diseases

The collaboration between the Critical Path Institute (C-Path) and the Engelberg Center for Health Care Reform at the Brookings Institution resulted in the Coalition Against Major Diseases (CAMD). This coalition aims to revolutionize the drug development process for neurodegenerative disorders, paving the way for addressing other critical diseases. The CAMD AD clinical trial dataset is a valuable asset for researchers committed to advancing AD studies, providing a wealth of information and insights in this crucial field. Through various statistical analysis techniques, researchers can conduct a comprehensive retrospective analysis of 5,952 patients enrolled in the placebo arms of 20 trials, offering valuable insights. This capability empowers researchers to explore the data and unlock valuable insights to combat AD. However, the pending public release of a data dictionary and data analysis syntax designed for statistical packages hampers the availability of necessary tools that could significantly improve the utilization of the dataset. The immense importance and complexity of this data necessitate streamlining the setup and analysis of the database to simplify the process for researchers, maximizing their ability to extract meaningful insights. The project aims to develop openly accessible syntax files to assist users of the Statistical Package for the Social Sciences (SPSS) in examining and analyzing the CAMD AD dataset. This endeavor simplifies the process, allowing users to gain valuable insights more effectively. By facilitating the creation and distribution of a standardized data dictionary and a

comprehensive set of syntax files, SPSS users will improve consistency and proficiency in conducting retrospective analyses of the CAMD AD datasets. This progress will, in turn, bolster more effective clinical trials of innovative AD therapies [161].

7.2.10 DementiaBank dataset

DementiaBank consists of recordings featuring 117 subjects diagnosed with AD and 93 healthy individuals. These recordings capture participants while they describe images. The information originates from a long-term research project focusing on AD and related forms of dementia. This study received funding from the National Institute on Aging and was carried out by the University of Pittsburgh School of Medicine. The dataset encompasses individuals diagnosed with MCI and various other types of dementia. Participants respond to questions in English based on their interpretation of the Cookie-Theft image, a component of the Boston Diagnostic Aphasia Examination (BDAE). This image from the BDAE is recognized for its clinical value in identifying linguistic deficits in individuals with AD and aphasia [162]. For additional information, readers can visit <https://dementia.talkbank.org/>.

7.2.11 Harvard aging brain study dataset

HABS is a long-term observational study focused on improving understanding of brain aging and the initial phases of AD. This investigation utilizes imaging data to identify early indicators of AD, including tau tangles and amyloid plaques, alongside comprehensive assessments of memory and other cognitive abilities using structural and functional imaging techniques. The primary objective is to furnish insights that accelerate progress toward effectively preventing cognitive decline associated with AD. The HABS website (<https://www.nmr.mgh.harvard.edu/lab/harvardagingbrain>) is the leading platform for data-sharing initiatives. The HABS dataset comprises fMRI tasks targeting both executive function and episodic memory. All participants do a Rule Switching Task (RST) at MRI visits. Extra tasks are administered to specific participants, including the Top-Down Suppression (TDS) task and a memory task incentivized with monetary rewards. Episodic memory tasks include Face-Name associative encoding, Face-Name Encoding Retrieval Flip (FNERF), and the Famous Face paradigm. Only the Face-Name associative encoding task is repeated at follow-up for about 60 participants [163].

7.2.12 Mayo clinic study of aging dataset

Mayo Clinic Study of Aging (MCSA) dataset is a population-based study investigating the prevalence, causes, and risk factors associated with MCI and dementia. It utilized the Rochester Epidemiology Project to identify the population of Olmsted County, Minnesota, aged 70–89, as of October 1, 2004. Randomly selected eligible individuals are invited to participate and undergo thorough in-person neuropsychological and neurological examinations. A consensus diagnosis of NC, MCI, and dementia was established by a panel using established criteria. A subset of participants underwent telephone interviews. The study assessed a total of 2,719 individuals, including 703 women (aged 70–79), 769 women (aged 80–89), 730 men (aged 70–79), and 517 men (aged 80–89). Among these, 2,050 individuals are assessed in person, while 669 are assessed via telephone. A comprehensive review of medical data identifies 402 individuals diagnosed with dementia [164]. After discussing datasets related to AD, the following section introduces a novel approach integrating holography microscopic medical imaging with DL for disease diagnosis. Table 3 summarizes the different datasets and databases associated with AD diagnosis.

8 Novel approach of integrating holography microscopic medical imaging with deep learning for diagnosis of diseases

This section of the paper introduces a novel, previously unexplored approach that utilizes holography microscopic medical imaging to diagnose diseases like AD. Holography microscopic imaging, inspired by physicist Dennis Gabor's work in 1948, combines holography and microscopy to see tiny biological samples [165]. "Holography" captures the complete three-dimensional structure of the object under consideration, while "microscopy" focuses on magnifying small objects. Unlike conventional microscopy techniques, holographic microscopy captures both the intensity and phase of light. The intensity component measures the amount of light, while the phase component details the light wave's position and direction. By recording both intensity and phase information, holographic microscopy can create intricate three-dimensional images, precisely examining cellular and sub-cellular structures. Holography microscopic imaging will offer researchers a unique and powerful tool for visualizing AD and its effects on the brain. By providing three-dimensional images with high spatial resolution, researchers will be able to examine the intricate architecture of brain tissues, including the distribution of amyloid plaques and tau tangles, in unprecedented detail. Further,

Table 3 Datasets with corresponding source links

Name of dataset	Description with source link
“Alzheimer’s Disease Neuroimaging Initiative (ADNI)” [151, 152]	The ADNI serves as a dynamic platform, uniting researchers and study data with a shared mission of gaining deeper insights into the progression of AD, thereby driving forward collective understanding of this debilitating condition. http://www.adni-info.org .
“Australian Imaging Bio-markers and Lifestyle(AIBL)” [153]	The AIBL research endeavors to precisely identify the biomarkers, cognitive attributes, lifestyle factors, and both physical and mental health aspects that significantly influence the progression of symptomatic AD. This concerted effort aims to uncover crucial insights into the development of the condition, advancing understanding in a profound manner. www.aibl.csiro.au
“Open Access Series of Imaging Studies (OASIS)” [150, 154]	With the objective of enabling early detection and identification of AD, the OASIS initiative strives to provide unrestricted access to brain neuroimaging datasets for researchers, fostering collaboration and empowering scientists to delve into the depths of this critical field. www.oasis-brains.org
“The National Alzheimer’s Coordinating Center (NACC) Database” [155]	Through fruitful collaboration with more than 42 AD Research Centers (ADRCs) across the United States, NACC has diligently curated and established an expansive and enduring Alzheimer’s database. This monumental resource stands as one of the largest and most influential worldwide, playing a crucial role in the collective efforts to combat this devastating disease. www.alz.washington.edu
“NIA genetics of Alzheimer’s disease data storage site (NIAGADS)” [156, 157]	With an unwavering commitment to advancing AD and associated dementias research, NIAGADS stands as an unwavering pillar, ensuring the standardization and widespread dissemination of genotypic data to the global scientific community. www.niagads.org
“The AD Knowledge Portal: A Repository for Multi-Omic Data on Alzheimer’s Disease and Aging” [158]	The AD Knowledge Portal hosts a publicly available dataset, housing a vast collection of data and resources. This valuable repository is the result of cooperative research projects focused on aging, dementia, and AD. Its accessibility drives knowledge exchange and fuels progress in the field of AD research. www.adknowledgeportal.synapse.org
“LORIS: a web based data management system for multi-center studies” [159]	LORIS, the Longitudinal Online Research and Imaging System, streamlines data management by integrating diverse study components and data types. It provides web-based accessibility, storage, processing, and sharing capabilities for imaging, clinical, behavioral, and genetics data. www.cbrain.mcgill.ca/loris
“The Minimal Interval Resonance Imaging in Alzheimer’s Disease (MIRIAD)” [160]	The MIRIAD dataset includes volumetric MRI brain scans from healthy seniors and Alzheimer’s patients. Its objective is to assess the viability of MRI as an outcome metric in AD clinical trials, driving advancements in research. http://miriad.drc.ion.ucl.ac.uk/
“Critical Path Institute (C-PATH) Online Data Repository (CODR): Coalition Against Major Diseases (CAMD) Alzheimer’s Disease Database” [161]	CODR, a public-private collaboration, hastens drug development for AD and other neurodegenerative disorders. Through advanced technologies, it aims to revolutionize treatment, restoring brain function with unparalleled speed and efficacy. https://c-path.org/
“DementiaBank Dataset” [162]	The DementiaBank dataset offers researchers audio and video recordings capturing interactions of individuals with dementia, notably Alzheimer’s, allowing studies on communication dynamics and cognitive deterioration. It is a valuable resource for advancing knowledge and strategies in managing neurodegenerative diseases. https://dementia.talkbank.org/
“Harvard Aging Brain Study (HABS) Dataset” [163]	The HAB Study dataset offers comprehensive data on aging and neurodegenerative conditions like Alzheimer’s, including demographics, clinical examinations, neuropsychological tests, and brain imaging, providing valuable insights into cognitive decline and early detection. https://www.nmr.mgh.harvard.edu/lab/harvardagingbrain

Table 3 (continued)

Name of dataset	Description with source link
“Mayo Clinic Study of Aging dataset (MCSA)” [164]	The Mayo Clinic Study of Aging dataset includes diverse examinations like neuropsychological tests, clinical evaluations, brain imaging, and biomarker data, offering insights into cognitive decline and neurodegenerative conditions. https://www.mayo.edu/research/centers-programs/alzheimers-disease-research-center/research-activities/mayo-clinic-study-aging/overview

the dynamic imaging capabilities of holography microscopy will enable the observation of real-time changes in brain tissue, aiding in studying AD progression and neuronal function. This comprehensive approach will allow researchers to identify early biomarkers of AD and track disease progression more effectively, ultimately facilitating the development of targeted therapeutic interventions.

In film-based holography, traditionally using photographic plates, holographic images are constructed by recording the interference pattern resulting from the interaction of light waves with the sample and a reference beam [166]. Holography microscopic imaging integration enables the capture of depth information, including quantitative data such as height and thickness. This approach provides an exhaustive understanding of the three-dimensional structure of biological samples, enhancing the analysis of microscopic features and phenomena. Based on the principles of refraction and diffraction, this procedure entails capturing the holographic image on light-sensitive film, which is then developed chemically. However, film-based techniques face drawbacks, such as time-consuming chemical processing, real-time viewing and other analysis constraints. In contrast, camera-based holography bypasses these challenges by utilizing digital cameras to directly capture the holographic interference pattern. This digital method enables instant viewing, analysis, and manipulation, improving flexibility and efficiency in current holographic imaging applications [167]. Figure 13 (a) and (b) demonstrates the traditional film-based holographic imaging procedure including hologram recording and reconstruction.

Several studies have investigated the application of holography microscopic imaging and AI techniques, such as ML and DL, for cancer diagnosis. This integration optimizes diagnostic efficiency and precision by utilizing holography's depth information and AI's analytical abilities. However, the integration of holography with DL methods remains relatively unexplored due to various challenges, such as the scarcity of publicly available datasets. A stain-free method introduced by [168] automatically categorizes live cancer cells using ML and Quantitative Phase Imaging (QPI). The technique analyzes spatial fluctuations in the cell phase, indicating stiffness and metastatic potential, employing holographic imaging for quantitative profiling during flow. They use SVM for cancer and blood cell classification, achieving 92.56% accuracy. Despite its potential for unbiased clinical diagnosis, the study needs to be improved

by a small training set (133 images per cell type), potentially impacting the ML classifier's performance and limiting generalizability. Furthermore, the focus on SW480 and SW620 cells restricts direct applicability to other cancer types or cell lines. A novel method for diagnosing Sickle Cell Disease (SCD) using 3D printed shearing digital holographic microscopy is introduced by [169]. Their research involves capturing video holograms of healthy and diseased Red Blood Cells (RBCs) using a compact interferometer, followed by manual segmentation and reconstruction of their optical path length profiles to estimate dynamic fluctuations. The model achieves greater accuracy in distinguishing between SCD-RBC and h-RBC (healthy-RBC) by using spatial and morphological features in conjunction with a random forest classifier.

A novel live-cell classification method using a triple-path DL framework integrating spatial and temporal fluctuation maps with quantitative optical thickness maps were introduced by [170]. They employ ResNet-50, which is pre-trained on ImageNet, and address gradient challenges using skip connections. The model trains independently on cell morphology and spatiotemporal maps, utilizing early and late fusion techniques to improve discrimination between primary and metastatic cells. Their architecture enhances the examination of metastatic potential compared to morphological evaluation. However, the study focuses on two specific cancer cell lines from a single patient, limiting generalizability to broader cancer types and diverse patient cohorts. [171] introduces an automated method for colorectal cancer screening using DL with SLIM (Spatial Light Interference Microscopy) data. They achieve high gland detection and classification accuracy by implementing a Mask R-CNN algorithm. Their study integrates a two-CONV-layer branch with a pre-trained ResNet101 from ImageNet, effectively detecting and classifying glands in SLIM images. This shows promise in colorectal tissue segmentation, classification, and whole-core diagnosis. However, the study's reliance on a tissue microarray with samples from only 132 patients may limit its representation of the diversity and variability of colorectal cancer cases within the broader population. Hence, additional validation and testing with wider and more diverse datasets are required to evaluate the applicability of their proposed method.

TOP-GAN, a pioneering DL method for medical imaging that seamlessly integrates transfer learning and GANs, was introduced by [172]. Their DL model achieves impressive

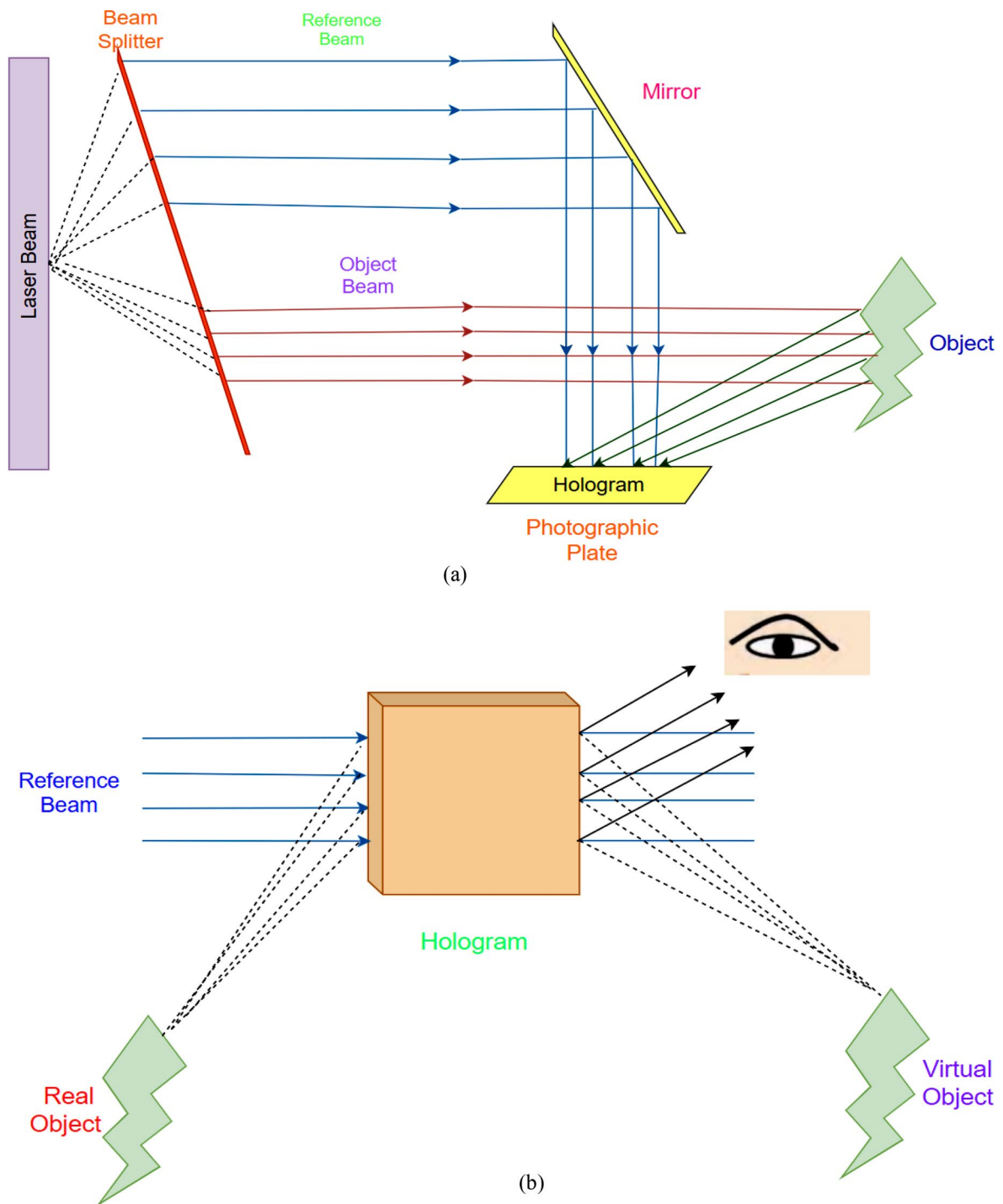


Fig. 13 Traditional film-based holographic imaging procedure. (a) Hologram recording. (b) Hologram reconstruction [177]

accuracy, ranging from 90 to 99%, in classifying healthy, primary, and metastatic cancer cells, even with a limited training dataset. TOP-GAN outperforms conventional methods in handling small training sets, showcasing the effectiveness of combining transfer learning and GANs. Their study suggests utilizing ImageNet's MOBILE-NET for optimal transfer learning with a small dataset, surpassing

alternatives such as VGG16. Although showing promising outcomes, the study recognizes constraints stemming from the small training set in DL. While the TOP-GAN approach exhibits promise, its advancement is hindered by the scarcity of classified images. Quantitative Transmitted Intensity Differential Interference Contrast (TI-DIC) microscopy and a DCNN by [173] contribute to diagnosing lung squamous

cell carcinoma. Their study utilizes a DCNN for classifying lung squamous cell carcinoma tissue based on DIC images or optical property maps, with the DCNN demonstrating superior accuracy, particularly with visual property maps. Incorporating label-free quantitative phase microscopy with DCNN shows potential for rapid and accurate cancer diagnosis. A study by [174] investigates the detection of Circulating Tumor Cells (CTCs) in blood samples obtained from cancer patients. The authors propose a staining-free

approach for CTC count using Digital Holographic Microscope (DHM), microfluidics, and ML. Utilizing holographic in-focus images, they developed S-Net, a specialized shallow network for classifying tumor cells in blood samples. S-Net outperformed ResNet-50, demonstrating superior accuracy, sensitivity, and specificity within shorter training periods, effectively minimizing false positives in mixed samples through an optimized threshold. Their study addresses the limitation of immunostaining for live CTC access, achieving successful tumor cell enumeration in lysed blood samples thereby paving the way for a stain-free method to count actual CTCs in cancer patients.

Based on our study analysis, no prior exploration of utilizing holography microscopic brain images for AD analysis with DL techniques has been conducted, highlighting a novel and potentially impactful area for research. Identifying this crucial and unidentified research gap will assist future researchers seeking to advance their careers in this field. Using holography microscopic brain imaging in AD diagnosis will be particularly beneficial due to its capacity to capture detailed three-dimensional brain structures. Unlike traditional imaging techniques like MRI, CT scan, etc., which primarily focus on amplitude (intensity) details; holography imaging will incorporate both amplitude and phase (angular) information. This will enable a more comprehensive examination of brain morphology and architecture, thereby improving the detection of minute changes indicative of early-stage AD pathology. By leveraging the depth and richness of three-dimensional imaging data, holography microscopic brain imaging will offer an exceptional means of identifying early disease markers compared to conventional imaging methods, thus advancing the potential for early diagnosis and intervention in AD. Table 4 presents a summarized overview of various DL studies utilizing holography imaging. After introducing the innovative approach of integrating holography microscopic medical imaging with DL for disease diagnosis, the following section delves into addressing challenges, identifying research gaps, and outlining future directions.

9 Challenges, research gaps and future directions

DL techniques hold promise for AD detection, yet they encounter significant challenges. This section discusses the challenges associated with identifying AD using DL models. Furthermore, the analysis thoroughly analyzes potential research gaps and identifies possible future directions in this field. Various identified challenges and, hence, future research directions are being discussed as follows:

Table 4 Holography microscopic imaging via DL

Reference	Integration Type= (Disease type + Imaging technique type + Kind of AI technique)	Evaluation metric
[168]	SW480 cells, Colorectal adenocarcinoma cells from colon tissue, and SW620 cells, metastatic cells from a lymph node of the same donor's colon, were studied + Holographic imaging + SVM	Accuracy (92.5%), Sensitivity: (88.88%), Specificity (96.25%), and AUC (0.961%) \
[169]	Macrophages (type of white blood cell) + Quantitative Phase Microscopy with transport of intensity equation + Multilevel integrated classifier and CLSNet with data augmentation	Accuracy ($93.1 \pm 2.4\%$), AUC ($98.5 \pm 1.7\%$) and Computational time (3.4 ± 0)
[170]	Live colorectal adenocarcinoma cancer cells + Digital off-axis holographic microscope + ResNet-50	Accuracy ($89.07 \pm 4.23\%$), Sensitivity ($88.89 \pm 7.50\%$), Specificity ($89.32 \pm 5.80\%$), Precision ($88.26 \pm 7.81\%$) and AUC ($96.03 \pm 2.44\%$)
[171]	Colorectal cancer + Spatial Light Interference Microscopy (SLIM) tissue scanner + Mask R-CNN	Accuracy (91% for gland detection, 99.71% in gland-level classification, and 97% Accuracy in core-level classification)
[172]	Two pairs of isogenic cell lines: (1) Hs 895.Sk (healthy skin) and Hs 895.T (melanoma), (2) SW 480 (colorectal adenocarcinoma colon cells) and SW 620 (metastatic from lymph node of colorectal adenocarcinoma cells) + Low-coherence off-axis holography + TOP-GAN	Sensitivity (98.96%) Specificity (99.61%) and AUC (99.5%)
[173]	Lung cancer + Quantitative Differential Interference Contrast Microscopy + Deep Convolutional Neural Network (DCNN)	AUC (96.8%), Accuracy (95.7%)
[174]	Breast cancer (MCF-7) and ovarian cancer (SkOV3) + Holography microscopy + S-Net (a custom-designed light-weight shallow network)	Accuracy (98.25%), Sensitivity (97.62%), and Specificity (98.89%)

- Limited availability of large, high-quality datasets:** The biggest hurdle in medical disease diagnosis is requirement of extensively, high-quality datasets. To give better results, DL models require large, diverse datasets rich in features to perform optimally. However, datasets are often susceptible to various data leakage issues, such as transfer learning bias and incorrect data splitting, which can skew model performance metrics. To address these challenges, meticulous dataset curation and validation processes are essential to ensure the reliability and generalizability of disease diagnosis models. Regardless, data gathering from hospitals can be complex due to ethical concerns and patient privacy issues, leading to limited availability of quality datasets. Collaborative agreements between institutions and hospitals are essential for this purpose and can thus prove beneficial for future researchers.
- Interpretability issues in DL models due to their black box nature:** The challenge of interpretability in DL models poses another significant hurdle in medical research problems analyzed using these models. In this context, understanding the underlying rationale behind DL model predictions impedes researchers' ability to provide explanations or justifications for their findings. This lack of transparency impedes progress in medical research, where clear insights and answers are essential for advancing knowledge and making informed decisions. Explainable AI techniques (like GradCAM and GradCAM++) aim to address AI models' lack of transparency and interpretability, allowing medical professionals to understand the rationale behind their decisions [175]. These techniques offer insights into how AI models make predictions by generating heatmaps that highlight cancerous or dangerous stages of the disease, thereby improving trust and confidence in their application. However, some challenges still need to be addressed. These include the complexity of specific AI models, which may require more advanced explanation methods and explainability techniques across different applications. For example, some complex models often involve numerous layers and parameters, making understanding how they arrive at their predictions difficult. More advanced explanation methods, such as layer-wise relevance propagation or saliency mapping, may be required to overcome this challenge. Future research can thus focus on improving the accuracy, relevance, and clarity of explanations given by AI models, an area that still requires further investigation.
- Class imbalance issue in disease classification:** The issue of class imbalance in medical datasets poses a substantial obstacle, mainly when specific disease categories are poorly represented. This complexity is heightened when considering the range of disease stages, from pre-diagnosis to progressive and severe irreversible phases. In such scenarios, the skewed distribution in the dataset can lead to biased model results, often neglecting the minority class, especially those representing severe or less prevalent disease stages. Therefore, model accuracy and reliability diminish, impeding their ability to deliver timely and precise diagnoses across all disease phases. Thus, developing practical strategies to tackle class imbalance and ensure equitable representation of all disease stages in the dataset is essential and challenging. Future researchers should explore advanced methods to address class imbalance, ensuring models can accurately predict all disease stages and improve diagnostic precision.
- Incorporation of holography microscopic brain imaging datasets for disease diagnosis as an open research challenge:** The central challenge identified in this research pertains to incorporating holography microscopic brain imaging datasets, which hold promise for improving early-stage disease diagnosis when combined with DL models. This recognized research gap symbolizes an open challenge that researchers still need to explore or acknowledge fully. Through this study's analysis, it becomes the first to highlight this open research challenge. The acquisition of this imaging data presents significant hurdles and requires extensive collaboration among institutes and hospitals to ensure dataset availability for future research.
- Use of advanced reinforcement learning algorithms for disease diagnosis:** While RL algorithms have shown promise in various domains, their direct application to AD diagnosis is currently limited. AD diagnosis typically involves analyzing medical imaging data, clinical assessments, and biomarkers, which may not align well with the sequential decision-making framework of RL. However, researchers are exploring the potential integration of RL techniques with other AI approaches for tasks related to treatment optimization, drug discovery, and personalized medicine in the context of AD. So, while RL algorithms may not be directly applicable to AD diagnosis, they could play a role in supporting related tasks in AD research and treatment. The utilization of RL represents a future research direction in this field, offering opportunities for innovation and advancement in AD diagnosis and treatment strategies.
- Difficulty in handling noisy or incomplete medical data, which can lead to suboptimal model performance:** Managing noisy or incomplete medical data is a notable challenge in developing precise disease diagnosis models. Such imperfections can significantly affect the performance of these models, restricting their ability

to make accurate predictions. Therefore, overcoming this hindrance entails implementing robust data preprocessing techniques to address noise and handle missing values effectively. Moreover, utilizing advanced AI algorithms capable of learning from imperfect data is vital. Successfully resolving these challenges is essential for improving the reliability and efficacy of medical diagnosis models, thereby improving patient results.

- **Challenge of data heterogeneity in medical research:** Data heterogeneity refers to the diversity or variability in data sources, formats, and acquisition protocols. In medical research problems, this manifests in various ways. Firstly, data originates from different sources such as hospitals, clinics, or research institutions, each utilizing its data collection practices and standards. Also, medical data exists in various formats, including structured data (e.g., electronic health records), unstructured data (e.g., medical images, clinical notes), and semi-structured data (e.g., laboratory reports). Moreover, differences in acquisition protocols, such as imaging parameters or clinical examination procedures, contribute to data heterogeneity. These variations pose challenges for DL models as they require standardized, homogeneous data to learn patterns and make accurate predictions effectively. Preprocessing techniques such as data normalization, standardization, and harmonization are necessary to ensure consistency and compatibility across datasets. This challenge is crucial for improving the performance and generalizability of DL models in medical research.
- **Cross-institutional data generalization challenge in DL models:** The domain adaptation challenge in medical research problems arises due to the variability in data distribution across different healthcare institutions. DL models trained on data from one institution may not generalize well to data from other institutions due to discrepancies in patient demographics, clinical practices, and imaging protocols. It presents a notable hindrance, limiting the applicability and efficacy of DL models across various healthcare environments. Therefore, handling domain adaptation challenges is essential for ensuring the reliability and generalizability of DL models across various healthcare institutions, ultimately improving patient care and clinical results. Future researchers can explore robust domain adaptation techniques critical for enhancing the effectiveness of DL models in diverse healthcare settings.
- **Ethical challenge in healthcare using AI:** Ethical considerations in healthcare using AI present multiple challenges, particularly concerning biases in training data and resulting predictions. Biases from demographic underrepresentation or skewed datasets may lead to unfair or inaccurate model results. Moreover, safeguarding

patient privacy and ensuring data security is critical due to the sensitive nature of medical information. Researchers can explore methods to strike a balance between harnessing AI for improved healthcare results and safeguarding patient confidentiality as a promising avenue for future research. Researchers can utilize federated learning [176] to address ethical concerns in healthcare AI. This approach allows collaborative model training across various devices or institutions, ensuring the privacy and security of patient data. Further, it helps mitigate biases by integrating diverse data sources, resulting in more accurate and unbiased AI models.

- **Challenge in acquiring longitudinal data for DL in disease diagnosis:** Collecting longitudinal data for diagnosing diseases like AD presents multiple obstacles. Firstly, it requires consistent and lengthy tracking of individuals' health status, which can be labor-intensive and resource-demanding. Moreover, the expenses of collecting, storing, and managing data can be significant. The ethical considerations, including patient privacy, consent, and data sharing, add another layer of complexity to this process. The effectiveness of longitudinal studies for disease diagnosis hinges on addressing these challenges through careful planning, the establishment of robust protocols, and appropriate resource allocation.

After discussing challenges, identifying research gaps, and outlining future directions, the following section moves into the discussion phase.

10 Discussion

AD is a severe illness in which brain cells break down over time, ultimately causing dementia. Dementia shows up as memory loss and trouble with communication. Dementia can show itself through a range of signs, including memory problems, challenges in communication, reasoning difficulties, personality alterations, struggles in day-to-day life activities, confusion about time or location, etc. The severity and progression of these symptoms can vary, affecting individuals' independence and quality of life. The incidence of AD is increasing globally, with significant trends observed in regions such as the United States. Based on statistical data, roughly 6.9 million individuals in the United States suffer from AD, with around two-thirds being women. The breakdown shows that there are 4.2 million women and 2.7 million men aged 65 and older with AD, representing 11% of women and 9% of men in this age group. Although there are treatment options for AD, they mainly focus on handling symptoms rather than curing or slowing down the disease's progression. Detecting and diagnosing AD early in

its progression is crucial as it facilitates timely interventions to slow its progression. Early disease identification could greatly benefit individuals by improving their chances of long-term survival rates. Therefore, identifying symptoms at an early stage plays a crucial role in managing and predicting AD. Advanced medical imaging techniques like MRI, CT scans, and PET scans offer valuable insights during the early stages of diagnosing diseases. Yet, the manual examination of these images can be time-consuming and error-prone. Moreover, minute symptoms crucial to disease progression might occasionally go unnoticed by doctors.

Researchers are noticing a significant increase in integrating ML, DL, and computer vision approaches (which are subsets of AI) with medical imaging techniques to address the limitations of traditional imaging modalities. This trend excels by the accessibility of publicly available large datasets, which DL models eagerly leverage to generate valuable insights and fruitful results. In literature, several research publications showcase researchers utilizing the analytical capabilities of DL models alongside medical imaging information like MRI, CT scans, PET, etc., to analyze and detect diseases earlier, thus improving patient survival rates and notably surpassing traditional imaging result analysis. This research study proposes a structured methodology that explores different imaging modalities, DL models, and challenges, such as data leakage issues, thereby identifying research gaps while offering insights into future research directions. The primary focus of this research paper revolves around analyzing various studies detecting AD at an early stage through integrating DL techniques with clinical data. It sheds light on the limitations of existing AD treatments, which only provide temporary relief from symptoms but do not provide a cure or stop the disease's progression. The paper critically evaluates relevant literature to investigate the potential synergies between DL and clinical data to improve early detection strategies. Various imaging methods, such as MRI, PET, CT, fMRI, and DTI, are crucial for AD diagnosis. MRI offers intricate brain structure images, PET identifies amyloid plaques and tau tangles, CT scans uncover brain irregularities, fMRI measures brain activity, and DTI estimates white matter integrity. According to the findings of this study, merging various imaging techniques with the analytical capabilities of DL models (like CNN, RNN etc.) can offer a comprehensive approach in understanding brain structure, function, and pathology in the early diagnosis of AD. Researchers in future can investigate how multimodal imaging techniques, including MRI, PET, and fMRI, could advance the early diagnosis of AD. This approach will provide a comprehensive examination of structural, functional, and molecular changes in the brain, enabling the detection of AD pathology at an earlier stage of symptoms. These initiatives may result in the formulation of

more precise diagnostic outcomes, ultimately allowing for earlier interventions with better results. The limited availability of high-quality, comprehensive imaging datasets poses significant challenge for researchers, hampering their ability to achieve expected research results. The initiation of collaborative efforts to improve data-sharing practices and prioritize the acquisition of comprehensive imaging datasets is critical for facilitating more impactful research initiatives in medicinal AI research.

This study has noted that while there are multiple traditional imaging techniques capable of providing three-dimensional images of brain structure, lacking in-depth detailed phase information. We identified that using advanced three-dimensional imaging modalities like holography microscopic medical imaging can prove beneficial for identifying intricate patterns by capturing both phase and amplitude information. However, the primary challenge in this approach is the need for publicly available holography microscopic imaging datasets. This identified gap can serve as a future research direction for researchers, highlighting the significance of incorporating amplitude and phase information in their analysis. A thorough examination of existing literature on holography imaging and DL has revealed significant research gaps and outlined potential future research avenues. The analysis underscores the significance of further investigating the integration of holography medical imaging and DL to improve diagnostic capabilities. However, there is an observation of a scarcity of studies that combine QPI and DL for AD diagnosis, with most studies focusing on specific cancer types rather than a broader spectrum. Also, much of the research tends to prioritize holography cell imaging over the integration of DL strategies. Therefore, addressing this research gap becomes a critical objective for future studies. Moreover, researchers can explore sophisticated image segmentation algorithms such as U-Net, DeepLab (Deep Labelling), Mask R-CNN, V-Net (Volumetric Neural Network), and Attention U-Net (Attention-based U-Net) to accurately identify brain structures and abnormalities for early diagnosis in AD. Moreover, the study has identified the importance of gathering longitudinal datasets for diagnosing AD, illustrating multiple benefits in tracking disease progression and understanding its progression. However, collecting longitudinal data for any disease poses challenges like maintaining participant engagement throughout the process, ensuring consistent data quality, addressing missing data, handling extensive data, and adhering to ethical guidelines. In medical research, emerging researchers may explore federated learning as a groundbreaking strategy to address ethical challenges, particularly in response to the multifaceted privacy concerns of medical datasets. Federated learning trains DL models across dispersed data sources, like individual devices or institutions.

Model updates are computed locally on each machine, and then results are communicated to a central server, consolidating them to update the overarching model. This iterative process repeats until the model achieves the desired performance while ensuring data privacy and security. Following an exhaustive literature review on AD detection utilizing DL techniques, it becomes apparent that specific unresolved challenges persist despite significant advancements in AI within this domain. These challenges involve difficulties related to the accessibility of comprehensive, high-quality datasets that are publicly unavailable, the lack of volumetric three-dimensional holography microscopic imaging datasets, and the recognition of federated learning as a promising direction for future research. After the discussion, the paper summarizes findings and insights in the concluding section.

11 Conclusion

Detecting AD at the earliest is crucial due to its progressive nature and the need to intervene before irreversible cognitive decline sets in. Although traditional medical examination techniques like MRI and CT scans are available, they are time-consuming and prone to errors, highlighting the need for more advanced techniques. Thankfully, advancements in AI-based methods offer a promising solution. By leveraging AI to analyze and process imaging data, it becomes possible to maximize the use of large datasets, extract crucial features, and identify intricate disease patterns. In this paper a critical review focused on the examination of the synergy between DL and clinical data in the early detection of AD is presented. The analysis highlights the substantial potential of DL algorithms in utilizing clinical data such as MRI, CT, PET, and advanced three-dimensional imaging modalities to improve the accuracy and efficiency of early-stage AD detection. By combining advanced computational techniques with comprehensive clinical information, researchers can harness the power of DL to enhance the predictive capabilities and diagnostic accuracy of AD detection models. Despite considerable research into the analytical capabilities of AI algorithms like ML and DL for medical image analysis, significant limitations persist, necessitating attention to crucial research areas. One such area, holography microscopic medical imaging for AD diagnosis, still needs to be explored and requires greater attention. Hence, this study proposes the integration of holographic imaging with DL as a groundbreaking advancement in AD detection. By providing three-dimensional visualization, holographic imaging can reveal intricate brain structures, capturing both amplitude and phase details. This transformative technology can potentially revolutionize the medical field and improve patient outcomes. Moreover, our investigation delves into

the data leakage problems in AD diagnosis, shedding light on how they can produce biased results. After evaluating the previous studies, it's clear that there's a critical necessity for a unified, automated solution that can accurately identify AD at its early progression toward the irreversible stage while minimizing the need for human interaction.

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Declarations

Ethical approval In compliance with ethical standards, this research, which does not involve human participants, adheres to established guidelines and principles, emphasizing the integrity and responsible conduct of the study.

Research Involving Human participants and/or animals informed consent This research does not involve the participation of human subjects and/or animals.

Informed consent This research does not involve the participation of human subjects and/or animals. Therefore, no such consent was required.

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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